MacroIR: a recursive likelihood framework for time-averaged stochastic data

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Abstract

Motivation: Many biological time series are recorded as temporal averages (e.g., macroscopic ionic currents), yet most inference methods assume instantaneous observations. **Results:** We introduce **MacroIR**, a recursive algorithm that computes the interval-averaged likelihood for Markovian systems by propagating the mean and covariance of ensemble state probabilities across measurement windows using an N^2 meta-state construction. **Validation:** MacroIR satisfies key information identities: $\mathbb{E}[\nabla_{\boldsymbol{\theta}} \log \mathcal{L}] = \mathbf{0}$ and $\mathrm{Var}(\nabla_{\boldsymbol{\theta}} \log \mathcal{L}) = \mathcal{I}(\boldsymbol{\theta})$, whereas non-recursive approximations violate them. **Impact:** This provides an internally testable likelihood for temporally averaged data. Code is provided.

1 Introduction

Ligand-gated ion channels (LGICs) are multimeric membrane proteins that convert chemical signals into electrical responses by coupling ligand binding to the opening of an ion-conducting pore [**? ? ? ? ?**]. By transforming the energy of ligand binding into conformational changes that modulate pore conductance, these receptors underpin fast synaptic transmission, muscle contraction, and other essential physiological processes [**? ? ?**]. Among LGICs, ATP-activated P2X receptors are particularly attractive as model systems due to their minimalist trimeric architecture, intersubunit ATP-binding sites [**?**], and the availability of high-resolution structures for both the closed [**?**] and open [**?**] states.

A comprehensive understanding of P2X receptor activation requires a framework that links the molecular mechanisms underlying the kinetic variability observed in experiments with the structural insights accessible via MD simulations. While single-channel Markov models capture the stochastic behavior of individual channels [**?**], macroscopic current fluctuations reflect the collective dynamics of many channels [**?**]. Historically, a disconnect has persisted between kinetic schemes derived from single-channel data and the molecular mechanisms governing channel activation [**?**]. The allosteric Monod–Wyman–Changeux (MWC) model [**?**] has provided a thermodynamic framework to reconcile these aspects, as demonstrated in studies on BK channels [**?**] and, more recently, purinergic receptors [**? ?**].

Here, we address this challenge with three complementary advances. First, we introduce *MacroIR*, a Bayesian framework that analyzes time-averaged macroscopic currents while preserving temporal correlations and enabling rapid evaluation of complex kinetic schemes. Unlike previous methods [**?**], *MacroIR* explicitly incorporates the system state at both the onset and conclusion of the averaging interval. Second, we develop conformational models that directly relate kinetic transitions to physical subunit rotations, eliminating the need for hypothetical intermediates (e.g., the flip state [? ? ?]). Third, we employ Molecular Dynamics (MD) simulations of single-bound P2X receptors to elucidate the asymmetry in subunit fluctuations predicted by our models.

Together, these advances provide robust Bayesian evidence and MD support for kinetic asymmetry in P2X2 receptor activation, reveal modulation of the activation barrier that prevents premature inactivation, and offer a general blueprint for integrating kinetic data with structural mechanisms.

2 Results

2.1 Interval-averaged likelihood via N^2 meta-states

We replace a system with N kinetic states by an equivalent meta-system with N^2 start–end states per interval. For each pair (i, j) we compute the expected interval-mean observable and its variance, yielding a Gaussian approximation at the ensemble level.

2.2 Recursive propagation and likelihood assembly

Let $\mathbf{x}_t$ denote the interval-averaged measurement at window t. MacroIR propagates $(\bar{\mathbf{p}}_t, \Sigma_{\mathbf{p},t})$ given $(\bar{\mathbf{p}}_{t-1}, \Sigma_{\mathbf{p},t-1})$ and transition rates $\boldsymbol{\theta}$, assembling the likelihood with interval-aware covariances.

2.3 Information identities: score–Fisher consistency

Define the total score $U(\boldsymbol{\theta}) = \sum_{t=1}^{T} \nabla_{\boldsymbol{\theta}} \log p(\mathbf{x}_t \mid \boldsymbol{\theta})$. Under correct specification, $\mathbb{E}[U] = \mathbf{0}$ and $\mathrm{Var}[U] = \mathcal{I}(\boldsymbol{\theta})$. We evaluate these by simulations at known $\boldsymbol{\theta}_0$ and show MacroIR satisfies them, while a non-recursive approximation does not.

2.4 Robustness across Δt, ensemble size, and noise

We map deviations as functions of window length Δt, number of channels N, and measurement noise, identifying regimes where Gaussian aggregation is tight.

3 Discussion

Our study bridges the longstanding gap between equilibrium-based allosteric models and phenomenological kinetic schemes by introducing a conformational kinetic framework that directly links discrete structural transitions to macroscopic channel function. By leveraging the MacroIR algorithm alongside detailed conformational modeling, we resolved multiple time constants associated with specific structural rearrangements during P2X2 receptor activation. This approach not only captures the evolution of state probabilities but also elucidates the relationship between partial activation and pore opening.

Our findings demonstrate that ATP binding induces an asymmetric conformational response among receptor subunits. Rather than triggering a uniform, concerted transition, ATP binding selectively lowers the energetic barrier for rotation in one subunit while leaving that of its neighbor largely unaltered. Moreover, the rotation of this favored subunit subsequently elevates the energetic barrier for ATP binding at an adjacent site, providing a mechanistic basis for the negative cooperativity observed in previous studies [?].

We identified the subunit with the reduced energetic barrier for rotation by analyzing single-bound P2X MD simulations. These simulations revealed a striking asymmetry in fluctuations along the rotation axis between the two chains forming the binding site. While chain B, on the right side of the binding site, exhibits only minor movements, chain A, on the left, shows considerably larger displacements, consistent with a lowered energetic barrier for rotation.

Although the activation kinetic profile could, in principle, be achieved without barrier modulation by assuming higher basal turnover rates, the differential modulation we observed suggests an increased energy barrier for resting-state rotation. This elevated barrier limits the time the channel spends in transitional conformations in the absence of an agonist, likely reducing the risk of spontaneous inactivation. Conversely, a reduction in both rotation and unrotation rates prolongs the time constant of unliganded gating noise, rendering it detectable by our analysis.

Mitigating risk by raising the energetic barrier in the inactive state appears to be a prudent strategy, leading us to propose that this mechanism constitutes a fundamental design principle in protein kinetics. By elevating the energy barriers for non-productive transitions, the receptor minimizes high-risk conformations. This strategy not only preserves protein functionality but also ensures rapid, ligand-induced responses when necessary. Overall, this mechanism, which fine-tunes transition frequency rather than merely shifting equilibrium populations, may represent a general strategy by which dynamic proteins balance functional responsiveness with resilience.

The flip state was initially proposed for glycine receptors [**?**] and later extended to purinergic receptors [**? ? ?**]. Here, we show that a model based on subunit rotation, rather than invoking a discrete flip state, can explain the same experimental data that originally motivated the flip state hypothesis for P2X2 receptors.

Beyond its mechanistic implications, our work has significant methodological and translational ramifications. The MacroIR framework provides a powerful tool for extracting kinetic information from time-averaged macroscopic currents, and its integration with structural data paves the way for a new era of dynamic structural biology. From a therapeutic standpoint, targeting kinetic couplings rather than equilibrium constants may yield next-generation modulators with improved specificity for conditions such as chronic pain and inflammation.

In summary, by unifying structural dynamics with kinetic principles, our study redefines allosteric regulation as an evolutionarily optimized modulation of transition pathways rather than mere state stabilization. This kinetic-conformational paradigm not only resolves long-standing discrepancies between structural snapshots and functional measurements but also opens new avenues for protein engineering and targeted drug design.

4 Methods

In this section, we detail our Bayesian analysis framework, including the experimental data used to evaluate our approach, the kinetic schemes developed for prediction, the likelihood function quantifying predictive accuracy, and the algorithm for computing Bayesian evidence to compare alternative kinetic models and infer posterior parameter distributions. In addition, we describe how the conformational ensemble of the closed P2X receptor—docked with an ATP molecule—was sampled using Molecular Dynamics simulations.

4.1 Bayesian Analysis

4.1.1 Why a Bayesian Approach?

A Bayesian framework is ideally suited for analyzing macroscopic currents because it directly addresses the challenge of temporal autocorrelation—a consequence of successive observations on the same finite population of channels in a macropatch. Both the MacroR and MacroIR algorithms exploit this autocorrelation to extract detailed kinetic information by continuously updating our belief about the probability distribution over the possible kinetic states.

At each stage, these algorithms iteratively refine the posterior probability distribution—approximated as a normal distribution—to yield a dynamic, high-resolution estimate of the kinetic states. Notably, this process also produces a correlation matrix among the states of different channels. It is important to emphasize that this correlation does not imply a causal influence between neighboring channels; rather, it arises because the aggregate current reflects the sum of individual channel contributions, thereby coupling the information on channel states in the observer's inference. In this sense, the phenomenon is reminiscent of quantum entanglement, where correlations become apparent only after measurement.

4.1.2 Objective of the Bayesian Analysis

The primary objective of our Bayesian analysis is to extract mechanistic insights from the stochastic fluctuations observed in macrocurrents. We achieve this through two complementary strategies:

1. **Comparison of Alternative Molecular Mechanisms and Kinetic Schemes.** We translate each proposed molecular mechanism into a corresponding kinetic scheme. In addition, we include previously proposed kinetic schemes where the underlying molecular mechanism remains ambiguous. Bayesian evidence for each alternative is computed using our MCMC framework, and the kinetic schemes that yield the highest evidence are interpreted as having the strongest support from the experimental data. It is important to note that this analysis does not prove that a particular molecular mechanism is occurring; rather, it facilitates selection among a set of plausible alternatives. Definitive validation is achievable only when the set of alternatives encompasses all logically possible options.

2. **Parameter Analysis of the Most Probable Kinetic Scheme.** In parallel with model comparison, we perform a detailed analysis of the parameters within the most probable kinetic scheme. In particular, the examination of kinetic coupling enables us to assess whether modulation of kinetic barriers occurs beyond simple regulation of equilibrium constants. Furthermore, evaluating the dependency of the current on the number of rotated subunits provides critical insight into how subunit rotation influences pore permeability.

Collectively, these approaches provide a robust framework for deriving mechanistic interpretations from macroscopic current recordings and for elucidating the kinetics of P2X2 receptor activation.

4.1.3 Likelihood Function and Model Parameters

The likelihood function defines the probability of observing the averaged recorded macroscopic current at each time interval, given a specific kinetic model and its associated parameters. The model parameters are categorized into distinct groups based on their functional role:

- **Kinetic Scheme Parameters:** Parameters that characterize the transitions between channel states, formulated either as state-based or allosteric schemes.

- **Channel Availability Model:** We modeled the evolution of the number of available channels over the course of the experiment by incorporating processes such as inactivation. In this model, two parameters are used: the total number of channels and the inactivation rate. Inactivation is represented as a distinct state that is irreversibly accessible from all other states at a uniform rate.

- **Conductance Model:** Parameters that define the single-channel conductance, including potential state-dependent variations.
- **Measurement Noise Model:** A model that accounts for stochastic fluctuations in the recorded signal, capturing both white and pink noise components.
- **Baseline Current Estimation:** A parameter that quantifies the residual baseline current present in the recordings.

All model parameters have a direct physical or material interpretation, meaning they correspond to measurable or experimentally relevant quantities. This ensures that parameter estimates can be directly related to the underlying biological and experimental conditions.

4.1.4 Prior Distribution

For each model, we specify a prior distribution for every parameter. Rather than encoding detailed prior knowledge of purinergic receptor kinetics, our priors were designed to encompass a broad range of plausible kinetic models, constrained only by general considerations. In this way, we aimed to favor kinetic schemes that structurally reflect receptor dynamics.

Since all parameters are inherently positive, we assigned each a $\log_{10}$-normal prior with a common variance of 2 in log-space, ensuring that the bulk of the probability mass spans roughly two orders of magnitude around the median. The medians of these priors were set to reasonable values, avoiding undue bias toward any particular model configuration. In cases where symmetry is present, such as when two parameters are interchangeable (e.g., right versus left coupling or in schemes 3 and 4, which incorporate two equivalent open states), one branch was assigned a median 10 times greater than its counterpart.

For the kinetic rate constants, we selected an intermediate value of 100 s^{-1}, while the association rate was set at 10 s^{-1} mM^{-1}. The equilibrium allosteric couplings were assigned the following values: BF = 100, BG = 1, FG = 100, RB = 100, BR = 10, and RBR = 1. By contrast, all kinetic allosteric couplings were uniformly set to 1.

Additional priors were specified as follows. The number of channels was fixed at 4800, the inactivation rate was set to 10^{-5} s^{-1}, and the measurement noise was defined with parameters of 10^{-3} pA·s for white noise and 5 pA for pink noise. The unitary current was set to 1 pA (measured at -60 mV). Moreover, the leakage current was fixed at 10^{-6} pA, the allosteric factor (which increases the current due to subunit rotation) was set to 100, and the basal current was specified as 1 pA.

Likelihood

The likelihood is defined below (Equation (**??**)). Note that MacroIR is an approximation, valid only within certain regimes. In our implementation, if the approximation is found to be invalid over a given interval, we employ the MacroINR algorithm for that interval. This logic is implemented in the function `Macror` (line 3576 of `qmodel.h`; `https://github.com/lmoffatt/macro_dr_submission`).

4.2 Experimental Data: Outside-out Patch Recording

The experimental dataset analyzed in this study was previously published in *The Journal of General Physiology* [**?**] and is reexamined here to develop and validate our approach. Recordings were obtained from macro outside-out patches containing hundreds to thousands of P2X2 channels. These patches were exposed to 0.2-ms ATP pulses at concentrations of 0.1, 0.2, 0.5, 1, 2, and 10 mM delivered every 2 minutes, interspersed with 10-ms pulses of 1 mM ATP. The dataset is ideally suited for kinetic analysis due to the precise alignment of stimulus timing with channel responses, which were recorded directly from the patch. For complete experimental details—including solution compositions, recording conditions, and data acquisition protocols—please refer to the original publication [**?**].

For kinetic analysis, we segmented the data as follows. The 1 s pre-pulse period was divided into 72 intervals of 13.7 ms and 10 intervals of 0.02 ms. During the ATP pulse, as well as 0.24 ms before, the algorithm processed the raw data, whereas after the pulse the data were averaged over exponentially spaced intervals (approximately 10 points per decade). Additionally, for each 10-ms ATP pulse, one averaged sample was retained from the second half of the pulse, corresponding to the equilibrium current.

4.3 Kinetic Models Under Test

We evaluated 11 kinetic models, including seven previously described by Moffatt and Hume [**?**] and four new models introduced in this study. These models can be classified into three distinct categories: **state models**, which describe receptor kinetics without explicitly incorporating allosteric interactions; **allosteric models**, which introduce modulations between specific transitions without an explicit structural correspondence; and **conformational models**, which explicitly map kinetic transitions onto physical structural rearrangements.

4.3.1 Previously Established Models

The seven previously established models represent progressively complex descriptions of receptor activation:

State Models: - *Scheme I*: A Minimal linear scheme for a homotrimeric channel in which all binding sites must be occupied before channel opening can occur. - *Scheme II*: Extends Scheme I by introducing an intermediate "flipped" state (a hypothesized intermediate conformational state preceding opening) between the fully bound and open conformations, allowing for a sequential activation process. - *Scheme III*: Preferred scheme on a single channel study [**?**], this scheme introduces multiple open states converging into a final closed state, refining receptor deactivation pathways. - *Scheme IV*: Builds upon Scheme III by incorporating a flipped state.

Allosteric Models: - *Scheme V*: A **Monod-Wyman-Changeux (MWC)** model for a trimeric receptor, in which ligand binding promotes a concerted gating transition through an allosteric mechanism. - *Scheme VI*: Introduces an explicit **flipping transition** that can be allosterically modulated. In this scheme, binding increases the probability of flipping, which in turn increases the probability of opening. - *Scheme VII*: Extends Scheme VI by allowing flipping to occur independently at each binding site, creating a highly interconnected state network in which binding, flipping, and gating are coupled.

At the time these models were developed, it was not yet known that the ATP-binding site of P2X receptors was inter-subunit, as in Cys-loop receptors, rather than intra-subunit, as in glutamate receptors. Consequently, these models did not account for the complexity of ternary coupling. Later, the resolution of closed and open structures of zP2X4 [**? ?**] confirmed the inter-subunit nature of the ATP-binding site and suggested that agonist binding promotes subunit rotation, hinting at a potential structural mechanism for gating.

A major limitation of the allosteric models (Schemes VI and VII) was that they not only postulated allosteric coupling between binding and flipping (and between flipping and gating)—which could be justified if flipping were the structural mechanism transmitting binding energy to the pore—but also required a direct binding–gating interaction to achieve a good fit. The latter was harder to interpret mechanistically.

4.3.2 New Conformational Models

Inspired by structural insights from P2X4, we developed four new models that replace the hypothetical "flipping" state with a well-defined **subunit partial rotation** and limit allosteric interactions to direct physical contacts—specifically, between ligand binding and subunit rotation. We considered three types of coupling based on the interacting elements:

- **RB (Rotation–Binding):** Coupling between the rotation of the subunit on the left of the ATP-binding site (subunit A in Figure **??**) and ATP binding.
- **BR (Binding–Rotation):** Coupling between the rotation of the subunit on the right of the ATP-binding site (subunit B in Figure **??**) and ATP binding.
- **RBR (Rotation–Binding–Rotation):** Ternary coupling that involves the rotation of both subunits forming the ATP-binding site in conjunction with ATP binding.

The four models are defined as follows:

- **Scheme VIII:** Incorporates only RBR coupling, where ATP binding induces a concerted rotation of both subunits that form the binding site.
- **Scheme IX:** Sets RBR to unity (i.e. no ternary coupling) and constrains RB equal to BR, thereby enforcing symmetric interactions between ligand binding and subunit rotations.
- **Scheme X:** Similar to Scheme IX, but allows RB and BR to vary independently, capturing potential asymmetries in the binding–rotation interactions.
- **Scheme XI:** Simultaneously incorporates all three coupling factors (RB, BR, and RBR), representing the most comprehensive model of receptor activation in this framework.

This conformational framework obviates the need for poorly defined allosteric interactions and provides a direct mapping between kinetic parameters and the structural changes observed in crystallographic studies.

4.3.3 Conductance Parameterization

For the conformational models (Schemes VIII, IX,X, XI), we hypothesized that conductance is a function of the number of rotated subunits. Instead of explicitly modeling the kinetics of a rotation–gating interaction (which would require a 48-state conformational model), we assumed that each rotated subunit stabilizes the open conformation, leading to an effective mean conductance that depends on the number of rotated subunits:

$$i(n) = i_{\max} \cdot \frac{E_n}{E_n + 1}, \quad E_n = E_0 \cdot F_g^n.$$

Here, E_n represents the efficacy of the n-rotated state. This formulation significantly reduced the number of states while maintaining sufficient flexibility to capture conductance modulation. In addition to computational feasibility, this approach naturally accounts for the possibility that the pore lacks a strict blocking mechanism, and that its conductance is a nonlinear function of the number of rotated subunits.

4.3.4 Kinetic and Equilibrium Allosteric Factors

Previous studies have extended the allosteric framework—typically applied to equilibrium constants—to kinetic rates [**?**]. In this work, we refine this approach by explicitly defining kinetic allosteric factors.

Consider two conformational transitions, g (gating) and r (rotation). In the absence of interaction, these transitions are characterized by four independent kinetic parameters: $b_{\text{on}}, b_{\text{off}}, r_{\text{on}}$, and r_{off}. When an allosteric interaction exists between binding and rotation (via either RB or BR coupling), the kinetic parameters in the activated state become

$$b^*_{\text{on}}, \quad b^*_{\text{off}}, \quad r^*_{\text{on}}, \quad r^*_{\text{off}}.$$

Microscopic reversibility imposes the following constraint on these parameters:

$$b_{\text{on}}\, r^*_{\text{on}}\, b^*_{\text{off}}\, r_{\text{off}} \;=\; r_{\text{on}}\, b^*_{\text{on}}\, r^*_{\text{off}}\, b_{\text{off}},$$

which can be rearranged as:

$$\frac{b_{\text{on}}/b_{\text{off}}}{b^*_{\text{on}}/b^*_{\text{off}}} \;=\; \frac{r_{\text{on}}/r_{\text{off}}}{r^*_{\text{on}}/r^*_{\text{off}}} \;=\; BR_{\text{eq}},$$

for BR coupling (a similar relation holds for RB coupling). Thus, only two extra parameters are required to specify the coupled kinetic rates, defined in terms of the forward rates:

$$BR_{b_{\text{on}}} \;=\; \frac{b^*_{\text{on}}}{b_{\text{on}}},$$

$$BR_{r_{\text{on}}} \;=\; \frac{r^*_{\text{on}}}{r_{\text{on}}},$$

with analogous definitions for RB coupling. For Bayesian model selection, we employed these forward-coupling parameters; however, for interpretative purposes, we also considered the corresponding reverse coupling factors:

$$BR_{b_{\text{off}}} \;=\; \frac{b^*_{\text{off}}}{b_{\text{off}}},$$

$$BR_{r_{\text{off}}} \;=\; \frac{r^*_{\text{off}}}{r_{\text{off}}}.$$

4.4 Likelihood function approximations of macroscopic currents

In this section, we introduce **MacroIR**, a novel algorithm for determining the likelihood of time-averaged macroscopic currents. MacroIR adapts the equations of our previously published macroscopic recursive algorithm (**MacroR**) [**?**] for calculating the posterior likelihood, repurposing them to operate on a *boundary state*—the composite state defined by the channel ensemble at both the beginning and the end of the time interval over which the current is averaged—instead of on the instantaneous state.

The rationale for focusing on the boundary state is rooted in the Markov property, which asserts that future dynamics depend solely on the present state rather than on the complete history. In our framework, the state at the beginning of an interval is sufficient to predict the dynamics within that interval, and the state at the end captures all the information necessary for forecasting the subsequent interval. To model the dynamics during the interval, we compute the mean of the averaged current conditioned on these boundary states and evaluate the variance of this mean, rather than the variance of the instantaneous current. This strategy directs our analysis to the overall signal properties, eschewing the transient fluctuations that occur within the interval.

4.4.1 Notation

We denote specific channel states using subscripts (e.g., i, j), while superscripts (e.g., "prior", "post", "obs") are used to indicate the type of probability distribution (e.g., prior, posterior, or observed) and we denote the intervals and the transitions from the initial state to the end state occur using arrows (e.g., $\rightarrow$) . For instance, p_i^{prior} represents the prior probability of being in state i, and p_i^{post} represents the posterior probability after an observation; $p_{i\rightarrow j}^{\text{prior}}$ and $p_{i\rightarrow j}^{\text{post}}$, on the other hand represents the prior and posterior probabilities of starting at the state i and ending at the state j.

4.4.2 Markov Model of Single-Channel Behavior

Ion channel gating is inherently stochastic. We model channel kinetics as a Markov process with a finite set of states k. The state occupancy is described by the probability vector $\boldsymbol{p}(t)$, which evolves according to

$$\frac{d\boldsymbol{p}}{dt}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{Q}, \tag{1}$$

where the off-diagonal elements of $\boldsymbol{Q}$ are the transition rates k_{ij} and the diagonal entries satisfy

$$k_{ii} = -\sum_{j\neq i} k_{ij}. \tag{2}$$

The solution is given by

$$\boldsymbol{p}(t) = \boldsymbol{p}(0) \cdot \exp(\boldsymbol{Q}t). \tag{3}$$

Since only the current generated by each state γ_i is measurable, the predicted current is

$$y^{\text{pred}}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{\gamma}. \tag{4}$$

This concise formulation links the hidden Markov dynamics to the observable current.

4.4.3 Calculation of the Average Current Conditional on Starting and Ending States

The time-averaged current for transitions from state i to state j over an interval t is defined as

$$\overline{\gamma}_{i\rightarrow j} = \frac{1}{t\, P_{i\rightarrow j}} \int_0^t \sum_k P_{i\rightarrow k}(\tau)\, \gamma_k\, P_{k\rightarrow j}(t-\tau)\, d\tau, \tag{5}$$

where $P_{i\rightarrow j}$ is the overall transition probability and γ_k is the current associated with state k.

Employing the spectral decomposition of the rate matrix $\mathbf{Q}$, this expression can be recast in closed form:

$$\overline{\gamma}_{i\rightarrow j} = \frac{1}{P_{i\rightarrow j}} \sum_{k,n_1,n_2} V_{in_1}\, V^{-1}_{n_1k}\, \gamma_k\, V_{kn_2}\, V^{-1}_{n_2j}\, E_2(\lambda_{n_1}t, \lambda_{n_2}t), \tag{6}$$

where $\mathbf{V}$ and $\mathbf{V}^{-1}$ are the eigenvector matrix of $\mathbf{Q}$ and its inverse, λ_n are the corresponding eigenvalues, and

$$E_2(x,y) = \begin{cases} \frac{e^x - e^y}{x-y}, & x \neq y, \\ e^x, & x = y. \end{cases} \tag{7}$$

4.4.4 Variance of the Average Current Conditional on Starting State

The variance of the time-averaged current for transitions starting at state i is defined as

$$\mathrm{Var}(\overline{\gamma}_{i\to .}) = E\big[\overline{\gamma}^2_{i\to .}\big] - \Big(E[\overline{\gamma}_{i\to .}]\Big)^2, \tag{8}$$

with $E[\overline{\gamma}_{i\to j}] = \sum_j \overline{\gamma}_{i\to j}\, P_{i\to j}$.

The expected square of the average current is initially written as

$$E\big[\overline{\gamma}^2_{i\to .}\big] = \frac{1}{t}\int_0^t \sum_{k_1,k_2} P_{i\to k_1}(\tau)\,\gamma_{k_1}\,P_{k_1\to k_2}(t-\tau)\,\gamma_{k_2}\,d\tau. \tag{9}$$

Using the spectral decomposition of $\mathbf{Q}$, a closed-form expression is obtained:

$$E\big[\overline{\gamma}^2_{i\to .}\big] = \sum_{k_1,k_2,n_1,n_2} V_{i\,n_1}\,V^{-1}_{n_1\,k_1}\,\gamma_{k_1}\,V_{k_1\,n_2}\,V^{-1}_{n_2\,k_2}\,\gamma_{k_2}, E_2(\lambda_{n_1}\,t, \lambda_{n_2}\,t), \tag{10}$$

where the auxiliary function E_2 is defined before as **??**

4.4.5 Variance of the Average Current Conditional on Starting and Ending States

The variance of the time-averaged current for transitions from state i to j is defined as

$$\mathrm{Var}(\overline{\gamma}_{i\to j}) = E\big[\overline{\gamma}^2_{i\to j}\big] - \Big(E[\overline{\gamma}_{i\to j}]\Big)^2, \tag{11}$$

with $E[\overline{\gamma}_{i\to j}]$ given by Eq. **??**.

The expected square of the average current is initially written as

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{1}{t^2\,P_{i\to j}}\int_0^t\int_0^{t-\tau_1} \sum_{k_1,k_2} P_{i\to k_1}(\tau_1)\,\gamma_{k_1}\,P_{k_1\to k_2}(\tau_2)\,\gamma_{k_2}\,P_{k_2\to j}(t-\tau_1-\tau_2)\,d\tau_1\,d\tau_2. \tag{12}$$

Using the spectral decomposition of $\mathbf{Q}$, a closed-form expression is obtained:

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{1}{P_{i\to j}} \sum_{k_1,k_2,n_1,n_2,n_3} V_{in_1}\,V^{-1}_{n_1k_1}\,\gamma_{k_1}\,V_{k_1n_2}\,V^{-1}_{n_2k_2}\,\gamma_{k_2}\,V_{k_2n_3}\,V^{-1}_{n_3j}\,E_3(\lambda_{n_1}t, \lambda_{n_2}t, \lambda_{n_3}t), \tag{13}$$

where the auxiliary function E_3 is defined as

$$E_3(x,y,z) = \begin{cases} E_{111}(x,y,z), & x\neq y,\ y\neq z,\ z\neq x, \\ E_{12}(x,y), & x\neq y,\ x\neq z,\ y=z, \\ E_{12}(y,z), & y\neq z,\ y\neq x,\ z=x, \\ E_{12}(z,x), & z\neq x,\ z\neq y,\ x=y, \\ \frac{1}{2}e^x, & x=y=z. \end{cases} \tag{14}$$

Here,

$$E_{12}(x,y) = E_{1,11}(x,y,y) + \frac{e^y}{y-x}\Big(\frac{y-x-1}{y-x}\Big), \tag{15}$$

and

$$E_{1,11}(x,y,z) = \frac{e^x}{(x-y)(x-z)}. \tag{16}$$

$$E_{111}(x,y,z) = \frac{e^x}{(x-y)\cdot(x-z)} + \frac{e^y}{(y-z)\cdot(y-x)}\,\frac{e^z}{(z-x)\cdot(z-y)} \tag{17}$$

4.4.6 From Single-Channel to Ensemble Channel Analysis

For an ensemble of ion channels, the state probability vector initially yields a multinomial distribution for the state counts. However, measuring the total current refines the posterior distribution and introduces inverse correlations among states generating the same current. A detailed but computationally intensive method—the *Microscopic Recursive Algorithm* [**?**]—constructs an ensemble state vector by applying Bayes' rule to every possible state count. For larger systems, we adopt a more efficient *Macroscopic Approach* that approximates the state distribution with a multivariate normal, using the covariance matrix to capture the observation-induced correlations.

4.4.7 MacroR: Macroscopic Bayesian Framework for Instantaneous Ensemble Measurements

Given the computational cost of the exact MicroR algorithm , we previously developed the *Macroscopic Recursive* (MacroR) approach as an efficient alternative for analyzing instantaneous measurements from channel ensembles [**?**]. MacroR operates on macroscopic variables: the mean state probability vector $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{p}]$ and the covariance matrix of state probabilities $\boldsymbol{\Sigma} = \mathrm{Cov}(\boldsymbol{p}, \boldsymbol{p})$, which describe the distribution of channels across states under a macroscopic approximation.

The core idea is to approximate the ensemble's state distribution with a multivariate normal distribution. This allows for a computationally tractable Bayesian update procedure. Within this framework, the likelihood of observing an instantaneous current y^{obs} is given by a Gaussian distribution:

$$\mathcal{L} = \mathcal{N}\Big(y^{\mathrm{obs}}; y^{\mathrm{pred}},\, \sigma^2_{y^{\mathrm{pred}}}\Big) \tag{18}$$

where y^{pred} is the predicted current based on the prior mean state probabilities:

$$y^{\mathrm{pred}} = N_{\mathrm{ch}}\,(\boldsymbol{\mu}^{\mathrm{prior}} \cdot \boldsymbol{\gamma}), \tag{19}$$

and the variance of the prediction, $\sigma^2_{y^{\mathrm{pred}}}$, incorporates multiple sources of variability:

$$\sigma^2_{y^{\mathrm{pred}}} = \epsilon^2 + \nu^2 + N_{\mathrm{ch}}\,(\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma}). \tag{20}$$

Here, ϵ^2 represents the variance of the instrumental white noise component. ν^2 represents a $\frac{1}{f}$ noise component (akin to 'pink noise' in its scaling with averaging time), intended to capture residual stochasticity not explicitly described by the kinetic model, such as model inaccuracies or contributions from unmodeled channel types. The final term, $N_{\mathrm{ch}}\,(\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})$, reflects the variance arising from the statistical distribution of the N_{ch} channels across different current-carrying states, as captured by the prior covariance matrix $\boldsymbol{\Sigma}^{\mathrm{prior}}$ within the multivariate normal approximation.

MacroR uses a Bayesian approach to update the estimated mean and covariance based on the observation. This approach was subsequently recognized as being mathematically equivalent to a Kalman filter formulation applied to this system [**?**]. The posterior mean and covariance are updated as follows:

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}^{\mathrm{prior}} + (y^{\mathrm{obs}} - y^{\mathrm{pred}})\,\frac{(\boldsymbol{\Sigma}^{\mathrm{prior}}\boldsymbol{\gamma})^{\mathrm{T}}}{\sigma^2_{y^{\mathrm{pred}}}}, \tag{21}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}^{\mathrm{prior}} - \frac{(\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})\,(\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})^{\mathrm{T}}}{\sigma^2_{y^{\mathrm{pred}}}}. \tag{22}$$

The prior estimates for the next time point $t_{n+1} = t_n + \Delta t_n$ are obtained by propagating the posterior estimates through the Markov process dynamics using the transition matrix $\mathbf{P}(\Delta t_n) = \exp(\boldsymbol{Q}\Delta t_n)$:

$$\boldsymbol{\mu}^{\mathrm{prior}}(t_{n+1}) = \boldsymbol{\mu}^{\mathrm{post}}(t_n)\,\mathbf{P}(\Delta t_n), \tag{23}$$

$$\mathbf{\Sigma}^{\text{prior}}(t_{n+1}) = \mathbf{P}(\Delta t_n)^{\text{T}}\Big(\mathbf{\Sigma}^{\text{post}}(t_n) - \text{diag}(\boldsymbol{\mu}^{\text{post}}(t_n))\Big)\mathbf{P}(\Delta t_n) + \text{diag}\Big(\boldsymbol{\mu}^{\text{prior}}(t_{n+1})\Big). \tag{24}$$

MacroR thus provides a recursive method to estimate the evolving state distribution of a channel ensemble from instantaneous measurements, forming a basis for the analysis of time-averaged data presented next with MacroIR.

4.4.8 MacroIR: A Bayesian Framework for Interval-Averaged Analysis of Channel Ensembles

We now introduce MacroIR (Macroscopic Interval Recursive), our Bayesian framework designed for analyzing sequences of time-averaged measurements from channel ensembles, such as those obtained in macropatch recordings over successive time intervals $[t_n, t_{n+1}]$. MacroIR extends the macroscopic approach by incorporating information about the system's dynamics *within* each averaging interval.

It leverages the concept of the **boundary state**—the joint state of starting the interval $[0, t]$ in state i_0 and ending in state i_t. The prior mean boundary state vector $\boldsymbol{\mu}^{\text{prior}}_{0\to t}$ and covariance matrix $\mathbf{\Sigma}^{\text{prior}}_{0\to t}$ characterizing this boundary state distribution are derived from the initial state distribution $(\boldsymbol{\mu}^{\text{prior}}_0, \mathbf{\Sigma}^{\text{prior}}_0)$ and the transition dynamics $\mathbf{P}_{0\to t} = \exp(\boldsymbol{Q}t)$:

$$(\mu^{\text{prior}}_{0\to t})_{(i_0\to i_t)} = (\mu^{\text{prior}}_0)_{i_0}\, P_{i_0\to i_t} \tag{25}$$

$$\begin{aligned}(\Sigma^{\text{prior}}_{0\to t})_{(i_0\to i_t)(j_0\to j_t)} = P_{i_0\to i_t}\Big((\Sigma^{\text{prior}}_0)_{i_0,j_0} - \delta_{i_0,j_0}\,(\mu^{\text{prior}}_0)_{i_0}\Big)P_{j_0\to j_t}\\ + \delta_{i_0,j_0}\,\delta_{i_t,j_t}\,(\mu^{\text{prior}}_0)_{i_0}\,P_{i_0\to i_t}\end{aligned} \tag{26}$$

A key feature of MacroIR is that the final likelihood calculation and the recursive propagation steps can be formulated without having to do the full expansion of the boundary state covariance matrix, $\mathbf{\Sigma}^{\text{post}}_{0\to t}$. This avoidance, arising from the algebraic simplification and marginalization of the Covariance Matrix during the update derivation, is crucial for computational efficiency compared to methods requiring explicit state-space expansion for boundary states.

The influence of the boundary path (start and end states) is captured through quantities like the mean current $\overline{\gamma}_{i\to j}$ and its variance $\sigma^2_{\overline{\gamma}_{i\to j}}$ conditional on the start and end states (defined in Sections **??** and **??**), which are pre-computed from the model kinetics and used in the likelihood.

The predicted average current over the interval $[0, t]$ depends on the initial state mean $\boldsymbol{\mu}^{\text{prior}}_0$ and the expected currents marginalized over the end states:

$$\overline{y}^{\text{pred}}_{0\to t} = N_{\text{ch}} \cdot (\boldsymbol{\mu}^{\text{prior}}_0 \cdot \overline{\boldsymbol{\gamma}}_0), \tag{27}$$

where $(\overline{\gamma}_0)_i = \sum_j P_{i\to j}(t)(\overline{\gamma})_{i\to j}$ represents the expected average current over the interval given the process started in state i. The variance of this prediction is:

$$\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}} = \epsilon^2_{0\to t} + N_{\text{ch}}\,(\gamma^{\text{T}}_{0\to t}\mathbf{\Sigma}_{0\to t}\gamma_{0\to t}) + N_{\text{ch}}\,(\boldsymbol{\mu}^{\text{prior}}_0 \cdot \boldsymbol{\sigma}^2_{\overline{\gamma}_0}), \tag{28}$$

with interval noise $\epsilon^2_{0\to t} = \epsilon^2/t + \nu^2$ (Eq. **??**) and the expected variance conditional on the start state $(\sigma^2_{\overline{\gamma}_0})_i = \sum_j P_{i\to j}\text{Var}(\overline{\gamma}_{i\to j})$. The scalar term $\gamma^{\text{T}}_{0\to t}\mathbf{\Sigma}_{0\to t}\gamma_{0\to t}$ represents the contribution to the predicted current's variance arising specifically from the uncertainty in the initial state distribution $(\mathbf{\Sigma}^{\text{prior}}_0)$. It is computed efficiently using the relationship:

$$\gamma^{\text{T}}_{0\to t}\mathbf{\Sigma}_{0\to t}\gamma_{0\to t} \;=\; \overline{\boldsymbol{\gamma}}^{\text{T}}_0\left(\mathbf{\Sigma}^{\text{prior}}_0 \;-\; \text{diag}(\boldsymbol{\mu}^{\text{prior}}_0)\right)\overline{\boldsymbol{\gamma}}_0 \;+\; \boldsymbol{\mu}^{\text{prior}}_0\left(\overline{\gamma}_{0\to t} \;\circ\; \overline{\gamma}_{0\to t} \;\circ\; \mathbf{P}_{0\to t}\right)\mathbf{1} \tag{29}$$

This calculation (Eq. **??**) bypasses the need to explicitly form $\mathbf{\Sigma}^{\text{prior}}_{0\to t}$, depending only on the initial state covariance $\mathbf{\Sigma}^{\text{prior}}_0$ and pre-calculated mean current terms.

The likelihood for observing the average current $\overline{y}^{\text{obs}}_{0\to t}$ over the interval is then:

$$\mathcal{L}_{0\to t} = \mathcal{N}\Big(\overline{y}^{\text{obs}}_{0\to t}; \overline{y}^{\text{pred}}_{0\to t},\, \sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}\Big). \tag{30}$$

Posterior updates can be efficiently computed without explicitly involving the meta-state. Specifically, we define the prior mean at time t by marginalization over the start state:

$$\boldsymbol{\mu}^{\text{prior}}_t = [\boldsymbol{\mu}^{\text{post}}_{0\to t}]_{.\to t} = \boldsymbol{\mu}^{\text{prior}}_0 \mathbf{P}(t) + \frac{\left(\overline{y}^{\text{obs}}_{0\to t} - \overline{y}^{\text{pred}}_{0\to t}\right)}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}} \left[\overline{\boldsymbol{\gamma}}^{\text{T}}_{0\to t}\, \boldsymbol{\Sigma_{0\to t}}\right]_{.\to t} \tag{31}$$

where the $. \to t$ subscript indicates the marginalization over the initial states i_0 of a vector V:

$$([V_{0\to t}]_{.\to t})_{i_t} = \sum_{i_0} V_{(i_0 \to i_t)} \tag{32}$$

where μ^{prior}_0 is the initial prior mean, $y^{\text{obs}}_{0\to t}$ and

$$\left[\overline{\boldsymbol{\gamma}}^T_{0\to t}\, \mathbf{\Sigma}^{\text{prior}}_{0\to t}\right]_{.\to t} = \overline{\boldsymbol{\gamma}}^T_0 \left[\mathbf{\Sigma}^{\text{prior}}_0 - \text{diag}\big(\boldsymbol{\mu}^{\text{prior}}_0\big)\right] \mathbf{P}(t) \;+\; \boldsymbol{\mu}^{\text{prior}}_0 \left(\overline{\boldsymbol{\gamma}}_{0\to t} \circ \mathbf{P}(t)\right). \tag{33}$$

The prior covariance for the next interval is calculated marginalizing the posterior over the start state , $\mathbf{\Sigma}^{\text{prior}}(t)$, incorporating the information gain from $\overline{y}^{\text{obs}}_{0\to t}$:

$$\begin{aligned}\mathbf{\Sigma}^{\text{prior}}_t = [\mathbf{\Sigma}^{\text{post}}_{0\to t}]_{.\to t} = \mathbf{P}^T_{0\to t}\Big(\mathbf{\Sigma}^{\text{prior}}_0 \;-\; \text{diag}\big(\boldsymbol{\mu}^{\text{prior}}_0\big)\Big)\, \mathbf{P}_{0\to t} \;+\; \text{diag}\big(\mu^{\text{prior}}_0 \cdot \mathbf{P}_{0\to t}\big) \\ -\; \frac{N_{ch}}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}} \left[\boldsymbol{\gamma}^T_{0\to t}\, \mathbf{\Sigma}^{\text{prior}}_{0\to t}\right]^{\text{T}}_{.\to t} [\boldsymbol{\gamma}^T_{0\to t}\, \mathbf{\Sigma}^{\text{prior}}_{0\to t}]_{.\to t}\end{aligned} \tag{34}$$

In comparison to other interval-based analysis methods like [**?**], MacroIR potentially offers higher accuracy. By incorporating the interval average centered around each time point for the updates, the state estimate at time t implicitly benefits from information regarding transitions influencing both the interval ending at t and the interval starting at t. This contrasts with methods applying only a single update based on the measurement within one interval [**?**]. While MacroIR involves more computations per time step to achieve this, its computational complexity scaling per step is expected to be similar to approaches like [**?**] that also avoid state-space expansion, because MacroIR efficiently bypasses the computation of the full posterior boundary state covariance.

In summary, MacroIR provides a recursive Bayesian framework tailored for interval-averaged measurements. By exploiting the mathematical structure to avoid explicit calculation of posterior boundary state covariances—using efficient computations involving marginalization over initial states (Eqs. **??**, **??**, **??**)—it computes likelihoods and propagates the state distribution, offering a potentially more accurate approach for analyzing macroscopic currents.

4.5 Likelihood function approximations of macroscopic currents

4.5.1 Markov Model of Single-Channel Behavior

Ion channel gating is inherently stochastic. We model channel kinetics as a Markov process with a finite set of states k. The state occupancy is described by the probability vector $\boldsymbol{p}(t)$, which evolves according to

$$\frac{d\boldsymbol{p}}{dt}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{Q}, \tag{35}$$

where the off-diagonal elements of $\boldsymbol{Q}$ are the transition rates k_{ij} and the diagonal entries satisfy

$$k_{ii} = -\sum_{j \neq i} k_{ij}. \tag{36}$$

The solution is given by

$$\boldsymbol{p}(t) = \boldsymbol{p}(0) \cdot \exp(\boldsymbol{Q}t). \tag{37}$$

Since only the current generated by each state γ_k (we use index k here to match Eq. **??**) is measurable, the predicted current is

$$y^{\text{pred}}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{\gamma}. \tag{38}$$

This concise formulation links the hidden Markov dynamics to the observable current.

4.5.2 Notation

We denote specific channel states using subscripts (e.g., i, j, k), while superscripts (e.g., "prior", "post", "obs") are used to indicate the type of probability distribution (e.g., prior, posterior, or observed) and we denote the intervals and the transitions from the initial state to the end state occur using arrows (e.g., $\rightarrow$) . For instance, p_i^{prior} represents the prior probability of being in state i, and p_i^{post} represents the posterior probability after an observation; $p_{i\rightarrow j}^{\text{prior}}$ and $p_{i\rightarrow j}^{\text{post}}$, on the other hand represents the prior and posterior probabilities of starting at the state i and ending at the state j. We use N_{ch} for the number of channels, $\boldsymbol{\mu}$ for the mean state probability vector, and $\boldsymbol{\Sigma}$ for the covariance matrix of the state probability distribution in macroscopic approximations.

4.5.3 From Single-Channel to Ensemble Channel Analysis

For an ensemble of N_{ch} ion channels, the state probability vector initially yields a multinomial distribution for the state counts $\boldsymbol{N} = (N_1, \ldots, N_k)$. However, measuring the total current y^{obs} refines the posterior distribution over $\boldsymbol{N}$ and introduces inverse correlations among states generating similar currents. A detailed but computationally intensive method—the *Microscopic Recursive Algorithm* (MicroR) [**?**]—constructs an ensemble state vector by applying Bayes' rule to every possible state count configuration $\boldsymbol{N}$. For larger systems ($N_{\text{ch}} \gg 1$), we adopt a more efficient *Macroscopic Approach* that approximates the state distribution with a multivariate normal distribution, characterized by its mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, which captures the observation-induced correlations.

4.5.4 Analysis of Instantaneous Current Measurements Overview

The following methods address the challenge of inferring the state of ion channels based on current measurements considered instantaneous relative to the channel gating dynamics. These approaches typically employ Bayesian inference, often recursively, to update the probability distribution over channel states or configurations upon observing the current at discrete time points. We first consider methods applicable to single channels (SingleR), then exact methods for microscopic ensembles (MicroR), and finally computationally tractable approximations for macroscopic ensembles (MacroR, MacroNR).

4.5.5 SingleR: Recursive Bayesian Inference for Single Channels

We introduce **SingleR**, a recursive method applied to individual (single-channel) states that redefines a well-established concept [?]. Although SingleR and MicroR share the same underlying algebra (Bayes' rule), they differ in state definitions: SingleR targets individual channel events, whereas MicroR addresses ensemble behavior ($\boldsymbol{N}$).

4.5.6 MicroR: Exact Bayesian Inference for Microscopic Ensembles

Experimental measurements often involve ensembles of N_{ch} identical channels. MicroR tracks the probability distribution over the specific number of channels occupying each state, the ensemble's meta-state $\boldsymbol{N} = (N_1, \ldots, N_k)$, subject to $\sum_i N_i = N_{\mathrm{ch}}$.

Given a prior distribution $P^{\mathrm{prior}}(\boldsymbol{N})$ and an observed instantaneous current y^{obs}, MicroR calculates the likelihood [?]:

$$\mathcal{L} = P(y^{\mathrm{obs}}) = \sum_{\boldsymbol{N}} P^{\mathrm{prior}}(\boldsymbol{N}) P(y^{\mathrm{obs}}|\boldsymbol{N}) \tag{39}$$

Assuming Gaussian instrumental noise ϵ^2, $P(y^{\mathrm{obs}}|\boldsymbol{N}) = \mathcal{N}(y^{\mathrm{obs}}; \sum_i N_i \gamma_i, \epsilon^2)$. Bayes' theorem updates the probability of each configuration:

$$P^{\mathrm{post}}(\boldsymbol{N}) = \frac{P^{\mathrm{prior}}(\boldsymbol{N}) P(y^{\mathrm{obs}}|\boldsymbol{N})}{P(y^{\mathrm{obs}})}. \tag{40}$$

Between measurements, $P(\boldsymbol{N}, t)$ evolves via the chemical master equation. MicroR is exact but computationally intractable for large N_{ch} and k due to the large number of configurations $\boldsymbol{N}$.

4.5.7 Macroscopic Approximations for Instantaneous Measurements: MacroR and MacroNR

To handle large ensembles, macroscopic methods approximate the ensemble state distribution, typically as multivariate normal characterized by mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$.

MacroR: Recursive Bayesian Framework for Instantaneous Measurements The *MacroR algorithm* [?] treats measurements as instantaneous and applies a recursive update analogous to a Kalman filter [?]. The likelihood is:

$$\mathcal{L} = \mathcal{N}\Big(y^{\mathrm{obs}}; y^{\mathrm{pred}}, \sigma^2_{y^{\mathrm{pred}}}\Big), \tag{41}$$

with predicted current $y^{\mathrm{pred}} = N_{\mathrm{ch}}\, \boldsymbol{\mu}^{\mathrm{prior}} \cdot \boldsymbol{\gamma}$ (Eq. ??) and variance $\sigma^2_{y^{\mathrm{pred}}} = \epsilon^2 + \nu^2 + N_{\mathrm{ch}}\, \boldsymbol{\gamma}^{\mathrm{T}}\, \boldsymbol{\Sigma}^{\mathrm{prior}}\, \boldsymbol{\gamma}$ (Eq. ??, assuming ν^2 represents other noise sources). The posterior mean and covariance are updated:

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}^{\mathrm{prior}} + \frac{y^{\mathrm{obs}} - y^{\mathrm{pred}}}{\sigma^2_{y^{\mathrm{pred}}}}\, N_{\mathrm{ch}}^{-1} \boldsymbol{\Sigma}^{\mathrm{prior}}\, \boldsymbol{\gamma}, \tag{42}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}^{\mathrm{prior}} - \frac{1}{\sigma^2_{y^{\mathrm{pred}}}} (\boldsymbol{\Sigma}^{\mathrm{prior}}\, \boldsymbol{\gamma})\, (\boldsymbol{\Sigma}^{\mathrm{prior}}\, \boldsymbol{\gamma})^{\mathrm{T}}. \tag{43}$$

Propagation uses $\mathbf{P}(\Delta t_n) = \exp(\boldsymbol{Q}\Delta t_n)$:

$$\boldsymbol{\mu}^{\mathrm{prior}}(t_{n+1}) = \boldsymbol{\mu}^{\mathrm{post}}(t_n)\, \mathbf{P}(\Delta t_n), \tag{44}$$

$$\boldsymbol{\Sigma}^{\mathrm{prior}}(t_{n+1}) = \mathbf{P}(\Delta t_n)^{\mathrm{T}} \Big(\boldsymbol{\Sigma}^{\mathrm{post}}(t_n) - \mathrm{diag}\big(\boldsymbol{\mu}^{\mathrm{post}}(t_n)\big)\Big) \mathbf{P}(\Delta t_n) + \mathrm{diag}\Big(\boldsymbol{\mu}^{\mathrm{prior}}(t_{n+1})\Big). \tag{45}$$

MacroNR: Non-Recursive Moment-Based Inference MacroNR [**? ?**] provides a non-iterative alternative, estimating likelihood without recursively updating the state distribution $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ after each measurement.

4.5.8 Analysis of Time-Averaged Measurements: Rationale and the Boundary State Concept

Experimental measurements are often averaged over a finite time interval $[t_n, t_{n+1}]$, yielding $\overline{y}^{\text{obs}}$. Analyzing such data requires methods accounting for dynamics *within* the interval. The **boundary state**—the joint probability distribution concerning the channel state at the beginning and end of the interval—is key. This leverages the Markov property: the start state predicts interval dynamics, the end state forecasts the next interval. We focus on the mean and variance of the *averaged* current conditioned on boundary states ($i \to j$), capturing overall signal properties.

4.5.9 Conditional Statistics for Boundary State Transitions

Analysis requires pre-calculating the expected current statistics during an interval t, conditional on starting in state i and ending in state j. Let $P_{i\to j} = P_{i\to j}(t) = (\exp(\boldsymbol{Q}t))_{ij}$.

Mean Conditional Current ($\overline{\gamma}_{i\to j}$) The time-averaged current for $i \to j$ transitions is:

$$\overline{\gamma}_{i\to j} = \frac{1}{t\, P_{i\to j}} \int_0^t \sum_k P_{i\to k}(\tau)\, \gamma_k\, P_{k\to j}(t-\tau)\, d\tau, \tag{46}$$

where γ_k is the current of state k. Using spectral decomposition $\mathbf{Q} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$:

$$\overline{\gamma}_{i\to j} = \frac{1}{P_{i\to j}} \sum_{k,n_1,n_2} V_{in_1}\, V^{-1}_{n_1k}\, \gamma_k\, V_{kn_2}\, V^{-1}_{n_2j}\, E_2(\lambda_{n_1}t, \lambda_{n_2}t), \tag{47}$$

where λ_n are eigenvalues and E_2 is defined as:

$$E_2(x,y) = \begin{cases} \frac{e^x - e^y}{x-y}, & x \neq y, \\ te^x, & x = y. \end{cases} \tag{48}$$

Variance of Conditional Current ($\mathrm{Var}(\overline{\gamma}_{i\to j})$) The variance is $\mathrm{Var}(\overline{\gamma}_{i\to j}) = E\big[\overline{\gamma}^2_{i\to j}\big] - (\overline{\gamma}_{i\to j})^2$ (Eq. **??**). The expected square involves a double integral:

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{2}{t^2\, P_{i\to j}} \int_0^t d\tau_1 \int_0^{\tau_1} d\tau_2 \sum_{k_1,k_2} P_{i\to k_1}(\tau_2)\, \gamma_{k_1}\, P_{k_1\to k_2}(\tau_1-\tau_2)\, \gamma_{k_2}\, P_{k_2\to j}(t-\tau_1). \tag{49}$$

The closed form involves eigenvalues and eigenvectors:

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{2}{P_{i\to j}} \sum_{k_1,k_2,n_1,n_2,n_3} V_{in_1}\, V^{-1}_{n_1k_1}\, \gamma_{k_1}\, V_{k_1n_2}\, V^{-1}_{n_2k_2}\, \gamma_{k_2}\, V_{k_2n_3}\, V^{-1}_{n_3j}\, E_3(\lambda_{n_1}t, \lambda_{n_2}t, \lambda_{n_3}t), \tag{50}$$

where E_3 involves auxiliary functions E_{111}, E_{12}:

$$E_3(x,y,z) = \begin{cases} E_{111}(x,y,z), & \text{distinct } x,y,z, \\ E_{12}(x,y), & y = z, x \neq y, \\ E_{12}(y,z), & z = x, y \neq z, \\ E_{12}(z,x), & x = y, z \neq x, \\ \frac{t^2}{2}e^x, & x = y = z. \end{cases} \tag{51}$$

$$E_{111}(x,y,z) = \frac{e^x}{(x-y)(x-z)} + \frac{e^y}{(y-z)(y-x)} + \frac{e^z}{(z-x)(z-y)}, \tag{52}$$

$$E_{12}(x,y) = \frac{e^x - e^y}{(x-y)^2} - \frac{te^y}{x-y}. \tag{53}$$

4.5.10 Microscopic Methods for Time-Averaged Data: SingleIR and MicroIR

To handle time-averaged data $\overline{y}^{\text{obs}}_{0\to t}$, SingleIR and MicroIR extend Bayesian analysis using the boundary state $s_{i\to j}$ (transition from state i at time 0 to state j at time t).

Assuming the mean current $\overline{\gamma}_{i\to j}$ follows a normal distribution (approximating the true distribution, justified if instrumental noise dominates gating noise), the conditional likelihood is:

$$P\left(\overline{y}^{\text{obs}} \mid s_{i\to j}\right) = \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \text{Var}(\overline{\gamma}_{i\to j}) + \epsilon^2_{0\to t}\right),$$

where $\epsilon^2_{0\to t} = \epsilon^2/t + \nu^2$ is the model-independent noise (Eq. **??**, assuming defined earlier or here).

The overall likelihood marginalizes over all boundary transitions:

$$\mathcal{L} = P\left(\overline{y}^{\text{obs}}\right) = \sum_{i,j} P^{\text{prior}}_{i\to j} \cdot \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \text{Var}(\overline{\gamma}_{i\to j}) + \epsilon^2_{0\to t}\right),$$

with prior $P^{\text{prior}}_{i\to j} = P^{\text{prior}}_i \cdot P_{i\to j}(t)$, where P^{prior}_i is the probability of being in state i at time 0. Bayes' rule gives the posterior probability of the transition:

$$P^{\text{post}}_{i\to j} = \frac{P^{\text{prior}}_{i\to j} \cdot \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \text{Var}(\overline{\gamma}_{i\to j}) + \epsilon^2_{0\to t}\right)}{P\left(\overline{y}^{\text{obs}}\right)}.$$

Marginalizing over the initial state i yields the prior probability $p^{\text{prior}}_j(t)$ for the start of the next interval:

$$p^{\text{prior}}_j(t) = \sum_i P^{\text{post}}_{i\to j}. \tag{54}$$

Cumulative log-likelihood is updated additively. SingleIR applies this to single channels, potentially addressing missing events or molecular motor analysis (though not demonstrated here). MicroIR applies this concept to microscopic ensembles, forming the conceptual basis for MacroIR.

4.5.11 MacroIR: A Bayesian Framework for Interval-Averaged Analysis of Channel Ensembles

MacroIR (Macroscopic Interval Recursive) analyzes time-averaged Markovian processes from ensembles [**?**] using the boundary state concept, characterized by prior mean $\boldsymbol{\mu}^{\text{prior}}_{0\to t}$ (Eq. **??**) and covariance $\boldsymbol{\Sigma}^{\text{prior}}_{0\to t}$ (Eq. **??**). Let $\mathbf{P} = \exp(\boldsymbol{Q}t)$.

The predicted average current uses the marginalized mean current $\overline{\boldsymbol{\gamma}}_0$ (where $(\overline{\boldsymbol{\gamma}}_0)_i = \sum_j \overline{\gamma}_{i\to j} P_{i\to j}$):

$$\overline{y}^{\text{pred}}_{0\to t} = N_{\text{ch}} \cdot \boldsymbol{\mu}^{\text{prior}}_0 \cdot \overline{\boldsymbol{\gamma}}_0. \tag{55}$$

The predicted variance involves interval noise $\epsilon^2_{0\to t}$, the expected variance $\boldsymbol{\sigma}^2_{\overline{\gamma}_0}$ (where $(\boldsymbol{\sigma}^2_{\overline{\gamma}_0})_i = \sum_j \text{Var}(\overline{\gamma}_{i\to j}) P_{i\to j}$), and the initial state uncertainty contribution $\widetilde{\gamma^{\text{T}}\boldsymbol{\Sigma}\gamma}$:

$$\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}} = \epsilon^2_{0\to t} + N_{\text{ch}} \cdot \widetilde{\gamma^{\text{T}}\boldsymbol{\Sigma}\gamma} + N_{\text{ch}} \cdot \boldsymbol{\mu}^{\text{prior}}_0 \cdot \boldsymbol{\sigma}^2_{\overline{\gamma}_0}. \tag{56}$$

$\widetilde{\gamma^{\mathrm{T}} \boldsymbol{\Sigma} \gamma}$ is computed efficiently:

$$\widetilde{\gamma^{\mathrm{T}} \boldsymbol{\Sigma} \gamma} = \overline{\boldsymbol{\gamma}}_0^{\mathrm{T}} \cdot \Big(\boldsymbol{\Sigma}_0^{\text{prior}} - \operatorname{diag}(\boldsymbol{\mu}_0^{\text{prior}}) \Big) \cdot \overline{\boldsymbol{\gamma}}_0 + \boldsymbol{\mu}_0^{\text{prior}} \cdot \Big((\overline{\boldsymbol{\gamma}}_{0\to t} \circ \mathbf{P}) \circ (\overline{\boldsymbol{\gamma}}_{0\to t} \circ \mathbf{P}) \Big) \cdot \mathbf{1}. \quad (57)$$

The likelihood is:

$$\mathcal{L}_{0\to t} = \mathcal{N}\Big(\overline{y}^{\text{obs}}_{0\to t}; \overline{y}^{\text{pred}}_{0\to t}, \, \sigma^2_{\overline{y}^{\text{pred}}_{0\to t}} \Big). \quad (58)$$

The formal posterior boundary state mean involves the marginalized gain term $\widetilde{\boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{\Sigma}} = \sum_{i_0} (\boldsymbol{\gamma}^{\mathrm{T}}_{0\to t} \cdot \boldsymbol{\Sigma}^{\text{prior}}_{0\to t})_{(i_0 \to \cdot)}$:

$$\boldsymbol{\mu}^{\text{post}}_{0\to t} = \boldsymbol{\mu}^{\text{prior}}_{0\to t} + \frac{\overline{y}^{\text{obs}}_{0\to t} - \overline{y}^{\text{pred}}_{0\to t}}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}} \cdot (\boldsymbol{\Sigma}^{\text{prior}}_{0\to t} \cdot \boldsymbol{\gamma}_{0\to t}). \quad (59)$$

The prior covariance for the next interval ($\boldsymbol{\Sigma}^{\text{prior}}(t)$) is updated using the propagated prior covariance and the information gain:

$$\boldsymbol{\Sigma}^{\text{prior}}(t) = \Big[\mathbf{P}^{\mathrm{T}} \Big(\boldsymbol{\Sigma}_0^{\text{prior}} - \operatorname{diag}(\boldsymbol{\mu}_0^{\text{prior}}) \Big) \mathbf{P} + \operatorname{diag}(\boldsymbol{\mu}^{\text{prior}}(t)) \Big] - \frac{1}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}} (\widetilde{\boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{\Sigma}})^{\mathrm{T}} \cdot (\widetilde{\boldsymbol{\gamma}^{\mathrm{T}} \boldsymbol{\Sigma}}). \quad (60)$$

MacroIR efficiently computes likelihoods and propagates distributions for interval data, avoiding explicit boundary state posterior covariance calculations.

4.6 Likelihood function approximations of macroscopic currents

4.6.1 MacroIR and Related Algorithms

In this section, we introduce **MacroIR**, a novel algorithm for determining the likelihood of time-averaged macroscopic currents. MacroIR adapts the equations of our previously published macroscopic recursive algorithm (**MacroR**) [**?**] for calculating the posterior likelihood, repurposing them to operate on a *boundary state*—the composite state defined by the channel ensemble at both the beginning and the end of the time interval over which the current is averaged—instead of on the instantaneous state.

The rationale for focusing on the boundary state is rooted in the Markov property, which asserts that future dynamics depend solely on the present state rather than on the complete history. In our framework, the state at the beginning of an interval is sufficient to predict the dynamics within that interval, and the state at the end captures all the information necessary for forecasting the subsequent interval. To model the dynamics during the interval, we compute the mean of the averaged current conditioned on these boundary states and evaluate the variance of this mean, rather than the variance of the instantaneous current. This strategy directs our analysis to the overall signal properties, eschewing the transient fluctuations that occur within the interval.

Previously, we introduced several algorithms for estimating the likelihood of instantaneous current measurements, including **MacroR** (macroscopic recursive), **MicroR** (microscopic recursive), and **MacroNR** (macroscopic non-recursive) [**?**]. MacroR iteratively updates the ensemble state of channels, akin to a Kalman filter, while MicroR applies Bayes' rule to ensembles of channels with identical kinetic properties to compute the exact likelihood. In addition, we introduce **SingleR**, a recursive method applied to individual (single-channel) states that redefines a well-established concept [?]. Although SingleR and MicroR share the same underlying algebra, they differ in state definitions: SingleR targets individual channel events, whereas MicroR addresses ensemble behavior. MacroNR, originally presented by Milescu [?] and later integrated into our framework [?], provides a non-iterative alternative by estimating the likelihood without updating the channel probability distribution after each measurement.

To extend our framework to time-interval data using the concept of boundary states, we define two complementary microscopic methods. Specifically, **MicroIR** forms the foundation of MacroIR by applying Bayesian inference to time-averaged measurements, while **SingleIR** offers an alternative approach designed to address the missing event problem and is proposed for applications to single channels or molecular motors (although detailed demonstrations of this application are not provided here). Collectively, these algorithms establish a comprehensive framework for analyzing current measurements across both microscopic and macroscopic scales.

4.6.2 Notation

We denote specific channel states using subscripts (e.g., i, j), while superscripts (e.g., "prior", "post", "obs") are used to indicate the type of probability distribution (e.g., prior, posterior, or observed) and we denote the intervals and the transitions from the initial state to the end state occur using arrows (e.g., $\rightarrow$) . For instance, p_i^{prior} represents the prior probability of being in state i, and p_i^{post} represents the posterior probability after an observation; $p_{i\rightarrow j}^{\text{prior}}$ and $p_{i\rightarrow j}^{\text{post}}$, on the other hand represents the prior and posterior probabilities of starting at the state i and ending at the state j.

4.6.3 Markov Model of Single-Channel Behavior

Ion channel gating is inherently stochastic. We model channel kinetics as a Markov process with a finite set of states k. The state occupancy is described by the probability vector $\boldsymbol{p}(t)$, which evolves according to

$$\frac{d\boldsymbol{p}}{dt}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{Q}, \tag{61}$$

where the off-diagonal elements of $\boldsymbol{Q}$ are the transition rates k_{ij} and the diagonal entries satisfy

$$k_{ii} = -\sum_{j\neq i} k_{ij}. \tag{62}$$

The solution is given by

$$\boldsymbol{p}(t) = \boldsymbol{p}(0) \cdot \exp(\boldsymbol{Q}t). \tag{63}$$

Since only the current generated by each state γ_i is measurable, the predicted current is

$$y^{\text{pred}}(t) = \boldsymbol{p}(t) \cdot \boldsymbol{\gamma}. \tag{64}$$

This concise formulation links the hidden Markov dynamics to the observable current.

4.6.4 MicroR: Microscopic Bayesian Inference for Ensemble State Counts

Experimental measurements often involve no single but ensembles of N_{ch} identical channels. A full microscopic treatment requires tracking the probability distribution over the specific number of channels occupying each state, known as the ensemble's meta-state or state configuration $\boldsymbol{N} = (N_1, N_2, \ldots, N_k)$, subject to the constraint $\sum_i N_i = N_{\text{ch}}$.

We previously introduced the *Microscopic Recursive* (MicroR) algorithm [**?**] to perform exact Bayesian inference on this ensemble state distribution using instantaneous current measurements. Given a prior probability distribution over configurations, $P^{\text{prior}}(\boldsymbol{N})$, and an observed instantaneous current y^{obs}, MicroR calculates the likelihood by summing over all possible configurations:

$$\mathcal{L} = P(y^{\text{obs}}) = \sum_{\boldsymbol{N}} P^{\text{prior}}(\boldsymbol{N}) P(y^{\text{obs}}|\boldsymbol{N}) \tag{65}$$

Here, $P(y^{\text{obs}}|\boldsymbol{N})$ represents the probability of observing y^{obs} given the ensemble is in configuration $\boldsymbol{N}$. Assuming Gaussian instrumental noise with variance ϵ^2, this is typically $\mathcal{N}(y^{\text{obs}}; \sum_i N_i \gamma_i, \epsilon^2)$, where $\sum_i N_i \gamma_i$ is the total current generated by configuration $\boldsymbol{N}$. Bayes' theorem is then used to update the probability of each configuration:

$$P^{\text{post}}(\boldsymbol{N}) = \frac{P^{\text{prior}}(\boldsymbol{N}) P(y^{\text{obs}}|\boldsymbol{N})}{P(y^{\text{obs}})}. \tag{66}$$

Between measurements, the probability distribution $P(\boldsymbol{N}, t)$ evolves according to the chemical master equation governing the ensemble dynamics.

While MicroR provides an exact solution, the number of distinct configurations $\boldsymbol{N}$ grows very rapidly with the number of channels N_{ch} and the number of states k. Consequently, MicroR becomes computationally intractable for typical macroscopic recordings. This computational barrier necessitates the use of approximations, such as the macroscopic moment-based approaches (MacroR and MacroIR) described subsequently, which avoid explicitly tracking the probability of every configuration.

4.6.5 SingleIR and MicroIR: Bayesian Inference for Time-Averaged Measurements

Ion channel currents cannot be observed instantaneously; instead, experiments measure the time-averaged current, denoted as $\overline{y}^{\text{obs}}_{0\to t}$, over a finite interval $[0, t]$. In this context, **SingleIR** and **MicroIR** extend classical Bayesian analysis for single channels and ensembles of channels to handle time-averaged measurements. Rather than applying Bayes' rule to instantaneous states, these methods use the *boundary state*—defined as the pair consisting of the channel state at the beginning and at the end of the integration interval—to condition the likelihood.

The time average of the current essentially represents the integrated current between the start and end of the interval. Given the state of the channel at the beginning and end of the interval, one can compute both the expected value (mean) $\overline{\gamma}_{i\to j}$ and the variance $\sigma^2_{\overline{\gamma}_{i\to j}}$ of the current for a transition from state i to state j. Under the assumption that the mean current follows a normal distribution, the conditional likelihood of observing $\overline{y}^{\text{obs}}$ given a boundary transition $s_{i\to j}$ is modeled as

$$P\left(\overline{y}^{\text{obs}} \mid s_{i\to j}\right) = \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \sigma^2_{\overline{\gamma}_{i\to j}} + \epsilon^2_{0\to t}\right),$$

where $\epsilon^2_{0\to t}$ accounts for model-independent noise and is decomposed into a white noise component ϵ^2 and a pink noise component ν^2, namely,

$$\epsilon^2_{0\to t} = \frac{\epsilon^2}{t} + \nu^2. \tag{67}$$

The overall likelihood for the time-averaged current is obtained by marginalizing over all possible boundary transitions:

$$\mathcal{L} = P\left(\overline{y}^{\text{obs}}\right) = \sum_{i,j} P^{\text{prior}}_{i\to j} \cdot \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \sigma^2_{\overline{\gamma}_{i\to j}} + \epsilon^2_{0\to t}\right),$$

with the prior probability for a transition given by

$$P_{i\to j}^{\text{prior}} = P_i^{\text{prior}} \cdot P_{i\to j},$$

and where the transition probability is

$$P_{i\to j} = P\big(s(t) = j \mid s(0) = i\big) = \exp\left(Q \cdot t\right).$$

Applying Bayes' rule yields the posterior probability of a transition from state i to j:

$$P_{i\to j}^{\text{post}} = \frac{P_{i\to j}^{\text{prior}} \cdot \mathcal{N}\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j}, \sigma^2_{\overline{\gamma}_{i\to j}} + \epsilon^2_{0\to t}\right)}{P\left(\overline{y}^{\text{obs}}\right)}.$$

Marginalization over the initial state yields the prior for the next interval:

$$p_j^{\text{prior}}(t_{n+1}) = \sum_i P_{i\to j}^{\text{post}}. \tag{68}$$

Cumulative log-likelihood is then updated additively over successive intervals. Although the distribution of the mean current is not strictly normal, we adopt this approximation under the assumption that instrumental noise predominates over gating noise.

This approach, while not implemented in the present work, establishes a conceptual framework for MacroIR and is readily generalizable to other single-molecule systems, such as molecular motors.

4.6.6 SingleIR and MicroIR: Microscopic Bayesian Inference for Time-Averaged Measurements

Instantaneous observations of ion channel currents are not technically possible, all experiments measure in fact time-averaged currents $\overline{y}^{\text{obs}}$ over some interval t. MicroIR extends classical Bayesian analysis of ensemble of channels to time-averaged measurements. El promedio temporal no es otra cosa que la integración entre dos tiempos, el inicio del intervalo y su finalización. MacroIR calcula la probabilidad de una serie de promedios temporales dado un modelo M con parámetros β a partir de una serie de productos de probabilidades condicionales:

$$P(y^{\text{obs}}_{t_i\to t_{i+1}}, y^{\text{obs}}_{t_{i+1}\to t_{i+2}}, \ldots |M(\beta)) = P(y^{\text{obs}}_{t_i\to t_{i+1}}|M(\beta)) \cdot P(y^{\text{obs}}_{t_{i+1}\to t_{i+2}}|y^{\text{obs}}_{t_i\to t_{i+1}}, M(\beta)) \cdot \ldots$$

Ahora, como consecuencia de la propiedad de Markov, todo lo que necesitamos para el período $t_{n+1} \to t_{n+2}$ es el estado al inicio de este período:

$$P(y^{\text{obs}}_{t_{n+1}\to t_{n+2}}|y^{\text{obs}}_{t_n\to t_{n+1}}, M(\beta)) = \sum_i P(y^{\text{obs}}_{t_{n+1}\to t_{n+2}} \mid s_{t_{n+1}} = i) \cdot P(s_{t_{n+1}} = i \mid y^{\text{obs}}_{t_n\to t_{n+1}}, M(\beta))$$

Entonces tenemos que obtener el posterior de $s_{t_{n+1}}$ dada el promedio temporal $y^{\text{obs}}_{t_n\to t_{n+1}}$ marginalizando para el estado al inicio del intervalo s_{t_n}.

De modo que lo unico importante es el conocimiento del estado del sistema en las boundaries de los intervalos, lo que ocurre durante los intervalos es promediado.

lo cual nos indica que como El meollo de la cuestión está en determinar la funcion recursiva La propiedad de Markov indica que para

Upon some reflection on the Markov property, it is clear that o

Although the channel state fluctuates continuously during t, Markov property indicates that all the information we need is the probability state at the beginning of the interval, the probability of the state trajectory can be determined using Equation **??** at the order to calculate the probability distribution of the averaged current over the successive intervals $t_i \to t_{i+1}$ and $t_{i+1} \to t_{i+2}$, we need not only the prior probability of the initial state i but also the end state $i+1$ to calculate the for the first interval and the final state ofand for calculating the probability of the next interval the end state i_2. MacroIR uses the Normal distribution as an approximation to the probability distribution of the average current. Therefore, it has to calculate the mean current during the interval during MicroIR relies solely on the initial state probability $\boldsymbol{p}^{\text{prior}}$ and its evolution via $\mathbf{P}(t)$ (Eq. **??**).

The joint probability of starting in state i and ending in state j is defined as

$$\Pi^{\text{prior}}_{i\to j} = p^{\text{prior}}_i \, P_{i\to j}, \tag{69}$$

or equivalently, in matrix form, $\mathbf{\Pi}^{\text{prior}} = \text{diag}(\boldsymbol{p}^{\text{prior}})\,\mathbf{P}$.

The likelihood of observing $\overline{y}^{\text{obs}}$ is given by

$$\mathcal{L} = \sum_{i,j} \Pi^{\text{prior}}_{i\to j}\, \mathcal{N}\!\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j},\, \epsilon^2_{0\to t} + \sigma^2_{\overline{\gamma}_{i\to j}}\right), \tag{70}$$

where $\Pi^{\text{prior}}_{i\to j}$ denotes the prior probability of a transition from state i to state j. The quantities $\overline{\gamma}_{i\to j}$ and $\sigma^2_{\overline{\gamma}_{i\to j}}$ represent the mean and variance, respectively, of the current associated with these transitions. The term $\epsilon^2_{0\to t}$ captures the variability not attributed to the model, which is decomposed into a white noise component ϵ^2 and a pink noise component ν^2

$$\epsilon^2_{0\to t} = \frac{\epsilon^2}{t} + \nu^2 \tag{71}$$

Using Bayes' rule, the posterior joint probability becomes

$$\Pi^{\text{post}}_{i\to j} = \frac{\Pi^{\text{prior}}_{i\to j}\, \mathcal{N}\!\left(\overline{y}^{\text{obs}}; \overline{\gamma}_{i\to j},\, \epsilon^2_{0\to t} + \sigma^2_{\overline{\gamma}_{i\to j}}\right)}{P(\overline{y}^{\text{obs}})}. \tag{72}$$

Marginalization over the initial state yields the prior for the next interval:

$$p^{\text{prior}}_j(t_{n+1}) = \sum_i \Pi^{\text{post}}_{i\to j}. \tag{73}$$

Cumulative log-likelihood is then updated additively over successive intervals. Although the exact distribution of the mean current is not strictly normal, we approximate it as such, assuming instrumental noise dominates over gating noise.

This approach, while not applied in the present work, provides a conceptual basis for MacroIR and can be generalized to other single-molecule systems, such as molecular motors.

4.6.7 Calculation of the Average Current Conditional on Starting and Ending States

The time-averaged current for transitions from state i to state j over an interval t is defined as

$$\overline{\gamma}_{i\to j} = \frac{1}{t\,P_{i\to j}} \int_0^t \sum_k P_{i\to k}(\tau)\,\gamma_k\,P_{k\to j}(t-\tau)\,d\tau, \tag{74}$$

where $P_{i\to j}$ is the overall transition probability and γ_k is the current associated with state k.

Employing the spectral decomposition of the rate matrix $\mathbf{Q}$, this expression can be recast in closed form:

$$\overline{\gamma}_{i\to j} = \frac{1}{P_{i\to j}} \sum_{k,n_1,n_2} V_{in_1}\, V^{-1}_{n_1 k}\, \gamma_k\, V_{kn_2}\, V^{-1}_{n_2 j}\, E_2(\lambda_{n_1} t, \lambda_{n_2} t), \tag{75}$$

where $\mathbf{V}$ and $\mathbf{V}^{-1}$ are the eigenvector matrix of $\mathbf{Q}$ and its inverse, λ_n are the corresponding eigenvalues, and

$$E_2(x,y) = \begin{cases} \frac{e^x - e^y}{x-y}, & x \neq y, \\ e^x, & x = y. \end{cases} \tag{76}$$

4.6.8 Variance of the Average Current Conditional on Starting and Ending States

The variance of the time-averaged current for transitions from state i to j is defined as

$$\mathrm{Var}(\overline{\gamma}_{i\to j}) = E\big[\overline{\gamma}^2_{i\to j}\big] - \Big(E[\overline{\gamma}_{i\to j}]\Big)^2, \tag{77}$$

with $E[\overline{\gamma}_{i\to j}]$ given by Eq. **??**.

The expected square of the average current is initially written as

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{1}{t^2\, P_{i\to j}} \int_0^t \int_0^{t-\tau_1} \sum_{k_1,k_2} P_{i\to k_1}(\tau_1)\, \gamma_{k_1}\, P_{k_1\to k_2}(\tau_2)\, \gamma_{k_2}\, P_{k_2\to j}(t-\tau_1-\tau_2)\, d\tau_1\, d\tau_2. \tag{78}$$

Using the spectral decomposition of $\mathbf{Q}$, a closed-form expression is obtained:

$$E\big[\overline{\gamma}^2_{i\to j}\big] = \frac{1}{P_{i\to j}} \sum_{k_1,k_2,n_1,n_2,n_3} V_{in_1}\, V^{-1}_{n_1 k_1}\, \gamma_{k_1}\, V_{k_1 n_2}\, V^{-1}_{n_2 k_2}\, \gamma_{k_2}\, V_{k_2 n_3}\, V^{-1}_{n_3 j}\, E_3(\lambda_{n_1} t, \lambda_{n_2} t, \lambda_{n_3} t), \tag{79}$$

where the auxiliary function E_3 is defined as

$$E_3(x,y,z) = \begin{cases} E_{111}(x,y,z), & x \neq y,\ y \neq z,\ z \neq x, \\ E_{12}(x,y), & x \neq y,\ x \neq z,\ y = z, \\ E_{12}(y,z), & y \neq z,\ y \neq x,\ z = x, \\ E_{12}(z,x), & z \neq x,\ z \neq y,\ x = y, \\ \frac{1}{2} e^x, & x = y = z. \end{cases} \tag{80}$$

Here,

$$E_{12}(x,y) = E_{1,11}(x,y,y) + \frac{e^y}{y-x}\Big(\frac{y-x-1}{y-x}\Big), \tag{81}$$

and

$$E_{1,11}(x,y,z) = \frac{e^x}{(x-y)(x-z)}. \tag{82}$$

$$E_{111}(x,y,z) = \frac{e^x}{(x-y)\cdot(x-z)} + \frac{e^y}{(y-z)\cdot(y-x)} \frac{e^z}{(z-x)\cdot(z-y)} \tag{83}$$

4.6.9 From Single-Channel to Ensemble Channel Analysis

For an ensemble of ion channels, the state probability vector initially yields a multinomial distribution for the state counts. However, measuring the total current refines the posterior distribution and introduces inverse correlations among states generating the same current. A detailed but computationally intensive method—the *Microscopic Recursive Algorithm* [**?**]—constructs an ensemble state vector by applying Bayes' rule to every possible state count. For larger systems, we adopt a more efficient *Macroscopic Approach* that approximates the state distribution with a multivariate normal, using the covariance matrix to capture the observation-induced correlations.

4.6.10 MacroR: Macroscopic Bayesian Framework for Instantaneous Ensemble Measurements

Given the computational cost of the exact MicroR algorithm (Section X.X.X), we previously developed the *Macroscopic Recursive* (MacroR) approach as an efficient alternative for analyzing instantaneous measurements from channel ensembles [**?**]. MacroR operates on macroscopic variables: the mean state probability vector $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{p}]$ and the covariance matrix of state probabilities $\boldsymbol{\Sigma} = \mathrm{Cov}(\boldsymbol{p}, \boldsymbol{p})$, which describe the distribution of channels across states under a macroscopic approximation.

The core idea is to approximate the ensemble's state distribution with a multivariate normal distribution. This allows for a computationally tractable Bayesian update procedure. Within this framework, the likelihood of observing an instantaneous current y^{obs} is given by a Gaussian distribution:

$$\mathcal{L} = \mathcal{N}\Big(y^{\mathrm{obs}}; y^{\mathrm{pred}},\, \sigma^2_{y^{\mathrm{pred}}}\Big) \tag{84}$$

where y^{pred} is the predicted current based on the prior mean state probabilities:

$$y^{\mathrm{pred}} = N_{\mathrm{ch}}\,(\boldsymbol{\mu}^{\mathrm{prior}} \cdot \boldsymbol{\gamma}), \tag{85}$$

and the variance of the prediction, $\sigma^2_{y^{\mathrm{pred}}}$, incorporates multiple sources of variability:

$$\sigma^2_{y^{\mathrm{pred}}} = \epsilon^2 + \nu^2 + N_{\mathrm{ch}}\,(\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma}). \tag{86}$$

Here, ϵ^2 represents the variance of the instrumental white noise component. ν^2 represents a $\frac{1}{f}$ noise component (akin to 'pink noise' in its scaling with averaging time), intended to capture residual stochasticity not explicitly described by the kinetic model, such as model inaccuracies or contributions from unmodeled channel types. The final term, $N_{\mathrm{ch}}\,(\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})$, reflects the variance arising from the statistical distribution of the N_{ch} channels across different current-carrying states, as captured by the prior covariance matrix $\boldsymbol{\Sigma}^{\mathrm{prior}}$ within the multivariate normal approximation.

MacroR uses a Bayesian approach to update the estimated mean and covariance based on the observation. This approach was subsequently recognized as being mathematically equivalent to a Kalman filter formulation applied to this system [**?**]. The posterior mean and covariance are updated as follows:

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}^{\mathrm{prior}} + (y^{\mathrm{obs}} - y^{\mathrm{pred}})\,\frac{(\boldsymbol{\Sigma}^{\mathrm{prior}}\boldsymbol{\gamma})^{\mathrm{T}}}{\sigma^2_{y^{\mathrm{pred}}}}, \tag{87}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}^{\mathrm{prior}} - \frac{(\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})\,(\boldsymbol{\Sigma}^{\mathrm{prior}}\,\boldsymbol{\gamma})^{\mathrm{T}}}{\sigma^2_{y^{\mathrm{pred}}}}. \tag{88}$$

The prior estimates for the next time point $t_{n+1} = t_n + \Delta t_n$ are obtained by propagating the posterior estimates through the Markov process dynamics using the transition matrix $\mathbf{P}(\Delta t_n) = \exp(\boldsymbol{Q}\Delta t_n)$:

$$\boldsymbol{\mu}^{\text{prior}}(t_{n+1}) = \boldsymbol{\mu}^{\text{post}}(t_n)\,\mathbf{P}(\Delta t_n), \tag{89}$$

$$\boldsymbol{\Sigma}^{\text{prior}}(t_{n+1}) = \mathbf{P}(\Delta t_n)^{\mathrm{T}}\Big(\boldsymbol{\Sigma}^{\text{post}}(t_n) - \operatorname{diag}(\boldsymbol{\mu}^{\text{post}}(t_n))\Big)\mathbf{P}(\Delta t_n) + \operatorname{diag}\Big(\boldsymbol{\mu}^{\text{prior}}(t_{n+1})\Big). \tag{90}$$

MacroR thus provides a recursive method to estimate the evolving state distribution of a channel ensemble from instantaneous measurements, forming a basis for the analysis of time-averaged data presented next with MacroIR.

4.6.11 MacroR: Bayesian Framework for Instantaneous Channel Ensemble Measurements

In the *MacroR algorithm*, we treat measurements as instantaneous, focusing on the ensemble's state via its mean probability vector and covariance matrix. This approach—equivalent to a Kalman filter[**?**]—yields the likelihood

$$\mathcal{L} = \mathcal{N}\Big(y^{\text{obs}} - y^{\text{pred}},\, \sigma^2_{y^{\text{pred}}}\Big), \tag{91}$$

with the predicted current

$$y^{\text{pred}} = N_{\text{ch}}\,\boldsymbol{\mu}^{\text{prior}}\,\boldsymbol{\gamma}, \tag{92}$$

and variance

$$\sigma^2_{y^{\text{pred}}} = \epsilon^2 + \nu^2 + N_{\text{ch}}\,\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\text{prior}}\,\boldsymbol{\gamma}. \tag{93}$$

The posterior mean and covariance are updated as

$$\boldsymbol{\mu}^{\text{post}} = \boldsymbol{\mu}^{\text{prior}} + \frac{y^{\text{obs}} - y^{\text{pred}}}{\sigma^2_{y^{\text{pred}}}}\,\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\text{prior}}, \tag{94}$$

$$\boldsymbol{\Sigma}^{\text{post}} = \boldsymbol{\Sigma}^{\text{prior}} - \frac{N}{\sigma^2_{y^{\text{pred}}}}\,\boldsymbol{\Sigma}^{\text{prior}}\,\boldsymbol{\gamma}\,\boldsymbol{\gamma}^{\mathrm{T}}\,\boldsymbol{\Sigma}^{\text{prior}}. \tag{95}$$

Propagation to the next time point is given by

$$\boldsymbol{\mu}^{\text{prior}}(t_{n+1}) = \boldsymbol{\mu}^{\text{post}}(t_n)\,\mathbf{P}(\Delta t_n), \tag{96}$$

$$\boldsymbol{\Sigma}^{\text{prior}}(t_{n+1}) = \operatorname{diag}\Big(\boldsymbol{\mu}^{\text{prior}}(t_{n+1})\Big) + \mathbf{P}(\Delta t_n)^{\mathrm{T}}\Big(\boldsymbol{\Sigma}^{\text{post}}(t_n) - \boldsymbol{\mu}^{\text{post}}(t_n)\Big)\mathbf{P}(\Delta t_n). \tag{97}$$

4.6.12 MacroIR: A Bayesian Framework for Interval-Averaged Analysis of Channel Ensembles

We now introduce MacroIR (Macroscopic Interval Recursive), our Bayesian framework designed for analyzing sequences of time-averaged measurements from channel ensembles, such as those obtained in macropatch recordings over successive time intervals $[t_n, t_{n+1}]$. MacroIR extends the macroscopic approach by incorporating information about the system's dynamics *within* each averaging interval.

It leverages the concept of the **boundary state**—the joint probability of starting the interval $[0, t]$ in state i_0 and ending in state i_t. The prior mean vector $\boldsymbol{\mu}^{\text{prior}}_{0\to t}$ and covariance matrix

$\mathbf{\Sigma}_{0\to t}^{\text{prior}}$ characterizing this boundary state distribution are derived from the initial state distribution $(\boldsymbol{\mu}_0^{\text{prior}}, \mathbf{\Sigma}_0^{\text{prior}})$ and the transition dynamics $\mathbf{P}(t) = \exp(\boldsymbol{Q}t)$:

$$(\mu_{0\to t}^{\text{prior}})_{(i_0\to i_t)} = (\mu_0^{\text{prior}})_{i_0}\, P_{i_0\to i_t}(t), \tag{98}$$

$$\begin{aligned}(\Sigma_{0\to t}^{\text{prior}})_{(i_0\to i_t)(j_0\to j_t)} = P_{i_0\to i_t}(t)\Big((\Sigma_0^{\text{prior}})_{i_0,j_0} - \delta_{i_0,j_0}\,(\mu_0^{\text{prior}})_{i_0}\Big)P_{j_0\to j_t}(t)\\ + \delta_{i_0,j_0}\,\delta_{i_t,j_t}\,(\mu_0^{\text{prior}})_{i_0}\,P_{i_0\to i_t}(t).\end{aligned} \tag{99}$$

A key feature of MacroIR is that, due to the structure of the Bayesian updates for this system, the final likelihood calculation and the recursive propagation steps are formulated to **avoid the explicit computation and storage of the full posterior boundary state covariance matrix**, $\mathbf{\Sigma}_{0\to t}^{\text{post}}$. This avoidance, arising from algebraic simplification and marginalization during the update derivation, is crucial for computational efficiency compared to methods requiring explicit state-space expansion for boundary states.

The influence of the boundary path (start and end states) is implicitly captured through quantities like the mean current $\overline{\gamma}_{i\to j}$ and its variance $\sigma^2_{\overline{\gamma}_{i\to j}}$ conditional on the start and end states (defined in Sections **??** and **??**)[1], which are pre-computed from the model kinetics and used in the likelihood.

The predicted average current over the interval $[0, t]$ depends on the initial state mean $\boldsymbol{\mu}_0^{\text{prior}}$ and the expected currents marginalized over the end states:

$$\overline{y}_{0\to t}^{\text{pred}} = N_{\text{ch}} \cdot (\boldsymbol{\mu}_0^{\text{prior}} \cdot \overline{\boldsymbol{\gamma}}_0), \tag{100}$$

where $(\overline{\gamma}_0)_i = \sum_j P_{i\to j}(t)(\overline{\gamma})_{i\to j}$ represents the expected average current over the interval given the process started in state i. The variance of this prediction is:

$$\sigma^2_{\overline{y}_{0\to t}^{\text{pred}}} = \epsilon^2_{0\to t} + N_{\text{ch}} \cdot \widetilde{\gamma^{\mathrm{T}}\mathbf{\Sigma}\gamma} + N_{\text{ch}} \cdot (\boldsymbol{\mu}_0^{\text{prior}} \cdot \boldsymbol{\sigma}^2_{\overline{\gamma}_0}), \tag{101}$$

with interval noise $\epsilon^2_{0\to t} = \epsilon^2/t + \nu^2$ (Eq. **??**) and the expected variance conditional on the start state $(\sigma^2_{\overline{\gamma}_0})_i = \sum_j P_{i\to j}(t)\mathrm{Var}(\overline{\gamma}_{i\to j})$. The term $\widetilde{\gamma^{\mathrm{T}}\mathbf{\Sigma}\gamma}$ represents the contribution to the predicted current's variance arising specifically from the uncertainty in the initial state distribution ($\mathbf{\Sigma}_0^{\text{prior}}$). While derived from operations involving the prior boundary state statistics (mean current vector $\boldsymbol{\gamma}_{0\to t}$ and covariance $\mathbf{\Sigma}_{0\to t}^{\text{prior}}$), it is computed efficiently using the relationship:

$$\begin{aligned}\widetilde{\gamma^{\mathrm{T}}\mathbf{\Sigma}\gamma} = \overline{\gamma}_0^{\mathrm{T}} \cdot \Big(\mathbf{\Sigma}_0^{\text{prior}} - \mathrm{diag}(\boldsymbol{\mu}_0^{\text{prior}})\Big) \cdot \overline{\gamma}_0\\ + \boldsymbol{\mu}_0^{\text{prior}} \cdot \Big((\overline{\gamma}_{0\to t} \circ \mathbf{P}(t)) \circ (\overline{\gamma}_{0\to t} \circ \mathbf{P}(t))\Big) \cdot \mathbf{1}.\end{aligned} \tag{102}$$

This calculation (Eq. **??**) bypasses the need to explicitly form $\mathbf{\Sigma}_{0\to t}^{\text{prior}}$, depending only on the initial state covariance $\mathbf{\Sigma}_0^{\text{prior}}$ and pre-calculated mean current terms.

The likelihood for observing the average current $\overline{y}_{0\to t}^{\text{obs}}$ over the interval is then:

$$\mathcal{L}_{0\to t} = \mathcal{N}\Big(\overline{y}_{0\to t}^{\text{obs}}; \overline{y}_{0\to t}^{\text{pred}},\, \sigma^2_{\overline{y}_{0\to t}^{\text{pred}}}\Big). \tag{103}$$

[1] Make sure these section labels exist or adjust references.

The Bayesian update uses the observation $\overline{y}^{\mathrm{obs}}_{0\to t}$ to refine the state estimate. The prior distribution for the *next* interval (starting at time t) is obtained efficiently. The key quantity driving the update is the vector $\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}}$, indexed by the end state i_t. This vector represents the covariance between the interval-averaged current and the probability of ending in state i_t. It is obtained by mathematically performing the matrix multiplication of the mean current vector associated with boundary transitions[2] ($\boldsymbol{\gamma}_{0\to t}$) with the prior boundary state covariance matrix ($\boldsymbol{\Sigma}^{\mathrm{prior}}_{0\to t}$), and then **marginalizing the result by summing over all possible initial states i_0**:

$$(\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}})_{i_t} = \sum_{i_0} \left(\boldsymbol{\gamma}_{0\to t} \cdot \boldsymbol{\Sigma}^{\mathrm{prior}}_{0\to t}\right)_{(i_0\to i_t)} = \sum_{i_0} \sum_{j_0, j_t} (\overline{\gamma})_{j_0\to j_t} (\Sigma^{\mathrm{prior}}_{0\to t})_{(j_0\to j_t)(i_0\to i_t)}. \tag{104}$$

This marginalized covariance vector $\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}}$ is then used to compute the prior covariance for the next interval, $\boldsymbol{\Sigma}^{\mathrm{prior}}(t)$, incorporating the information gain from $\overline{y}^{\mathrm{obs}}_{0\to t}$:

$$\boldsymbol{\Sigma}^{\mathrm{prior}}(t) = \left[\mathbf{P}(t)^{\mathrm{T}}\left(\boldsymbol{\Sigma}^{\mathrm{prior}}_0 - \mathrm{diag}(\boldsymbol{\mu}^{\mathrm{prior}}_0)\right)\mathbf{P}(t) + \mathrm{diag}(\boldsymbol{\mu}^{\mathrm{prior}}(t))\right]_{\text{Propagated Covariance}} - \frac{1}{\sigma^2_{\overline{y}^{\mathrm{pred}}_{0\to t}}} (\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}})^{\mathrm{T}} \cdot (\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}}). \tag{105}$$

The prior mean for the next interval, $\boldsymbol{\mu}^{\mathrm{prior}}(t)$, is similarly updated (details omitted for brevity, but involves $\widetilde{\boldsymbol{\gamma}^{\mathrm{T}}\boldsymbol{\Sigma}}$ and marginalization of the formal posterior boundary state mean given by Eq. **??**).

In comparison to other interval-based analysis methods like [**?**], MacroIR potentially offers higher accuracy. By incorporating the interval average centered around each time point for the updates, the state estimate at time t implicitly benefits from information regarding transitions influencing both the interval ending at t and the interval starting at t. This contrasts with methods applying only a single update based on the measurement within one interval [**?**]. While MacroIR involves more computations per time step to achieve this, its computational complexity scaling per step is expected to be similar to approaches like [**?**] that also avoid state-space expansion, because MacroIR efficiently bypasses the computation of the full posterior boundary state covariance.

In summary, MacroIR provides a recursive Bayesian framework tailored for interval-averaged measurements. By exploiting the mathematical structure to avoid explicit calculation of posterior boundary state covariances—using efficient computations involving marginalization over initial states (Eqs. **??**, **??**, **??**)—it computes likelihoods and propagates the state distribution, offering a potentially more accurate approach for analyzing macroscopic currents.

We introduce MacroIR (Macroscopic Interval Recursive), a Bayesian framework for analyzing time-averaged Markovian processes over successive intervals, as encountered in macropatch recordings. In contrast with previous time-averaging algorithms [**?**], MacroIR leverages the **boundary state**, which represents the joint probability of starting in state i_0 and ending in state i_t over an interval $[0, t]$.

The boundary state prior mean is defined as:

$$(\mu^{\mathrm{prior}}_{0\to t})_{(i_0\to i_t)} = (\mu^{\mathrm{prior}})_{i_0}\, P_{i_0\to i_t}(t), \tag{106}$$

with the corresponding covariance:

$$(\Sigma^{\mathrm{prior}}_{0\to t})_{(i_0\to i_t)(j_0\to j_t)} = P_{i_0\to i_t}\left((\Sigma^{\mathrm{prior}}_0)_{i_0,j_0} - \delta_{i_0,j_0}\,(\mu^{\mathrm{prior}}_0)_{i_0}\right)P_{j_0\to j_t} + \delta_{i_0,j_0}\,\delta_{i_t,j_t}\,(\mu^{\mathrm{prior}}_0)_{i_0}\,P_{i_0\to i_t} \tag{107}$$

[2]Let $\boldsymbol{\gamma}_{0\to t}$ denote the vector of mean currents $\overline{\gamma}_{j_0\to j_t}$, indexed by boundary pairs $(j_0 \to j_t)$.

The predicted average current over the interval is given by

$$\overline{y}^{\text{pred}}_{0\to t} = N_{\text{ch}} \cdot \mu_0^{\text{prior}} \cdot \overline{\gamma}_0, \tag{108}$$

where the marginalized current per state is

$$(\overline{\gamma}_0)_i = \sum_j (\overline{\gamma})_{i\to j}. \tag{109}$$

The variance of the predicted current is:

$$\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}} = \epsilon^2_{0\to t} + N_{\text{ch}} \cdot \widetilde{\gamma^{\text{T}}\boldsymbol{\Sigma}\gamma} + N_{\text{ch}} \cdot \mu_0^{\text{prior}} \cdot \sigma^2_{\overline{\gamma}_0}, \tag{110}$$

with

$$(\sigma^2_{\overline{\gamma}_0})_i = \sum_j (\sigma^2\,\overline{\gamma})_{i\to j}. \tag{111}$$

The effective term $\widetilde{\gamma^{\text{T}}\boldsymbol{\Sigma}\gamma}$ is computed as:

$$\begin{aligned}\widetilde{\gamma^{\text{T}}\boldsymbol{\Sigma}\gamma} = \gamma^{\text{T}}_{0\to t} \cdot \boldsymbol{\Sigma}^{\text{prior}}_{0\to t} \cdot \gamma_{0\to t} = &\\ \overline{\gamma}_0^{\text{T}} \cdot \Big(\boldsymbol{\Sigma}_0^{\text{prior}} - \operatorname{diag}(\mu_0^{\text{prior}})\Big) \cdot \overline{\gamma}_{0\to t}&\\ + \mu_0^{\text{prior}} \cdot \Big(\overline{\gamma}_{0\to t} \circ \overline{\gamma}_{0\to t} \circ \mathbf{P}\Big) \cdot \mathbf{1},&\end{aligned} \tag{112}$$

where $\circ$ denotes the element-wise (Hadamard) product and $\mathbf{1}$ is a vector of ones.

The likelihood function for the time interval $0 \to t$ is therefore

$$\mathcal{L}_{0\to t} = \mathcal{N}\Big(\overline{y}^{\text{obs}}_{0\to t} - \overline{y}^{\text{pred}}_{0\to t},\, \sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}\Big) \tag{113}$$

Posterior updates are performed without explicitly computing the boundary state. The posterior mean is updated as:

$$\mu^{\text{post}}_{0\to t} = \mu^{\text{prior}}_{0\to t} + \frac{y^{\text{obs}}_{0\to t} - y^{\text{pred}}_{0\to t}}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}} \cdot \widetilde{\boldsymbol{\gamma}^{\text{T}}\boldsymbol{\Sigma}}, \tag{114}$$

with

$$\widetilde{\boldsymbol{\gamma}^{\text{T}}\boldsymbol{\Sigma}} = \mathbf{1}_0^{\text{T}} \cdot \gamma^{\text{T}}_{0\to t} \cdot \boldsymbol{\Sigma}^{\text{prior}}_{0\to t}. \tag{115}$$

Finally, the prior covariance for the next interval is updated by marginalizing over the initial states:

$$\begin{aligned}\boldsymbol{\Sigma}_t^{\text{prior}} = \mathbf{P}^{\text{T}}\Big(\boldsymbol{\Sigma}_0^{\text{prior}} - \operatorname{diag}(\boldsymbol{\mu}_0^{\text{prior}})\Big)\mathbf{P} + \operatorname{diag}(\boldsymbol{\mu}_0^{\text{prior}} \cdot \mathbf{P})&\\ - \frac{N}{\sigma^2_{\overline{y}^{\text{pred}}_{0\to t}}}\, \widetilde{\boldsymbol{\gamma}^{\text{T}}\boldsymbol{\Sigma}}^{\text{T}} \cdot \widetilde{\boldsymbol{\gamma}^{\text{T}}\boldsymbol{\Sigma}}.&\end{aligned} \tag{116}$$

In summary, MacroIR provides a recursive Bayesian framework for interval-averaged measurements, efficiently computing likelihoods and posterior distributions while circumventing the computational burden of explicit boundary state calculations.

4.7 Bayesian Model Comparison

We compared 11 alternative kinetic schemes to evaluate distinct mechanistic hypotheses for P2X2 receptor activation. For each scheme, a fully specified Bayesian model was constructed with explicit likelihood functions and prior distributions as described above. Bayesian evidence was estimated using a modified parallel-tempering, affine-invariant[**?**]Markov chain Monte Carlo (MCMC) algorithm with dynamic temperature selection [**?**]. Rather than enforcing a uniform exchange rate between neighboring temperature chains, our implementation aims to maintain a uniform product of $\Delta \log \mathcal{L}$ and $\Delta\beta$. This strategy equalizes exchange rates at high β, while reducing the frequency of extremely negative $\Delta\beta \cdot \log \mathcal{L}$ values at low β, thereby obviating the need to justify their exclusion. The algorithm efficiently sampled high-dimensional posterior distributions by dynamically adapting the temperature configuration during sampling, which minimized potential sampling biases and ensured a uniform error rate in log-evidence estimation across temperatures.

4.7.1 MCMC Convergence and Resolution

MCMC convergence was assessed by monitoring the potential scale reduction factor (PSRF) for each parameter and derived quantity, ensuring that effective sample sizes (ESS) were sufficiently large. Detailed PSRF and ESS statistics, along with credibility intervals for log-evidence estimates, the number of iterations, computational time, and system specifications (number of cores and processor used), are provided in Supplementary Table 1. Supplementary Table 2 examines the convergence of parameter estimates for Scheme 10 using the MacroIR approach, while Supplementary Table 3 presents parameter estimates for $\beta = 0$, validating our implementation of the affine-invariant ensemble Monte Carlo Markov Chain by confirming the expected median and variance of the prior distributions.

4.7.2 Validation of MacroIR Using Simulated Data

To assess the robustness of our Bayesian framework, we validated the MacroIR method using simulated recordings. Synthetic ATP-activated currents were generated based on the fitted kinetic parameters for Scheme X, ensuring that simulation conditions closely matched the experimental setup. We then applied MacroIR to this dataset to evaluate its ability to recover the original parameters. As shown in Supplementary Table 4, all but two parameters were successfully recovered within their respective credibility intervals. The two outliers were the estimated Number of Channels (overestimated) and the Single-Channel Current (underestimated).

These two parameters primarily influence stochastic fluctuations without affecting time kinetics. A larger estimated channel count reduces fluctuations through random averaging, leading to a compensatory underestimation of single-channel conductance to maintain the same overall current. Since MacroIR is an approximation, some degree of bias is expected. Here, the overestimation of the Number of Channels suggests a form of overfitting, where an underestimation of stochastic fluctuations leads to a compensatory overestimation of kinetic components. Nevertheless, with only two outliers among 16 parameters—each with credibility intervals covering 90% of the total probability—these deviations fall within expected variability but warrant further investigation.

4.7.3 Posterior Predictive Checks

To evaluate model fit, we performed posterior predictive checks. In the MacroIR approach, each measurement is modeled as a normal distribution with a predicted mean current and variance. These predicted distributions were compared to the experimental data by computing a log-likelihood measure; the agreement between the observed and expected log-likelihood values confirms that the model accurately reproduces the experimental data.

4.8 Conformational characterization of the closed form of P2X perturbed by the binding of an ATP in one binding site

4.8.1 Models for Molecular Dynamics simulations

P2X receptors are homomeric or heteromeric assemblies of three monomers labeled $P2X_1$ through $P2X_7$. Up to date, Protein Data Bank includes experimental structures from distinct species of P2X1 [?], P2X3 [? ? ? ? ?], P2X4 [? ? ? ?], and P2X7 [? ? ? ? ? ?]. Most of the structures lack complete intracellular residues, which are crucial for conducting stable MD simulations and preventing channel drift [?]. Besides, when AlphaFold [?] is implemented to predict these regions, results display very low confidence.

P2X receptors vary in the extent of their permeation selectivity [?]. However their overall architecture —particularly the extracellular region— is conserved across subtypes and species. Each channel resembles a chalice with each monomer having a dolphin-like shape. The extracellular domain is relatively large with binding sites located in the upper part, approximately 40 $\AA$ away from the membrane surface. This conserved structural design strongly implies a unified closed→open pathway triggered by ATP. Only P2X4 and P2X7 were determined in both their closed and open states P2X4 [? ? ? ? ?]. According to our experience in practicing reliable MD simulations [? ? ? ?], and due to the common opening mechanism of the distinct subtypes, we examined the perturbations caused by ATP docking to a single binding site considering three distinct models based on zebrafish $P2X_4$ (Uniprot code: F8W463).

Two of these models implied the open ($P2X^{open}$) and closed ($P2X^{closed}$) form of the channel. These structures were used to define a vector that describes the changes in the extracellular vestibule between both conformations. This region presents the largest differences in the transition $P2X^{closed} \rightarrow P2X^{open}$ [?] and coupled to the ion conducting transmembrane domain (See Fig. ??). The models were built considering Protein Data Bank: codes 3I5D (closed) and 4DW1 (open). The following paragraphs explain the modifications made to the original PDB files to equalize the number of residues of both forms. In PDB code 3I5D, chain-A spans from Arg36 to Ile359, chain-B from Leu34 to Leu361, and chain-C from Leu34 to Trp358. To standardize the lengths, we added residues Leu34, Asn35, Val360, and Leu361 to chain-A and residues 359-361 to chain-C, using chain-B as a reference. PDB code 4DW1, by the other side, includes only one of the three monomers and it includes the sequence Arg36-Ile359. We supplemented this chain with residues Leu34, Asn35, 360, and 361 to match the total number of residues in the closed structure. The atomic coordinates for the other two chains were derived using symmetry information from the crystallographic data. Additionally, we incorporated residues Val31, Gly32, and Thr33 into each chain of both structures. The conformation of these inserted segments was predicted using AlphaFold3 [?], which suggested that the new residues would continue the α-helix motif of the previous residues. In this way, the final forms of $P2X^{closed}$ and $P2X^{open}$ contains residues from Val31 to Leu361 in each of their three chains. $P2X^{open}$, besides, contains the coordinates of an ATP molecule in each of its three binding site.

The third model was constructed to investigate, through Molecular Dynamics (MD) simulations, the initial conformational changes caused by the interaction of an ATP molecule to the binding site conformed by chains A and B. Chain A refers to the one that contacts ATP via its head and left flipper subdomains (marked in blue in Fig. ??), while chain B is the one that does it through its dorsal fin (coloured in red in Fig. ??). To construct this model, an ATP molecule was docked into the binding site of $P2X^{closed}$ formed by chains A and B. The ATP coordinates were determined by aligning the Cα atom positions of $P2X^{open}$ to those of $P2X^{closed}$. This resulting model will be referred as $P2X^{closed}_{1-ATP}$.

4.8.2 Vector definition for the close $\longrightarrow$ open transition of the extracellular vestibule

It has been described that the opening of the transmembrane channel involves an outward 'flexing' of the lower region of the extracellular vestibule (EV) [**?**]. In order to measure how the binding of ATP affects the EV dynamical behaviour, we defined a vector for each of the two subunits that composed the binding site where the ATP was docked in $P2X^{closed}_{1-ATP}$. To construct these vectors, $P2X^{open}$ and $P2X^{closed}$ were first aligned and then the coordinates of Cα atoms from residues Asp59-Ser65, Asn195-Lys205, Arg261-Pro275, and Phe327-Phe333 of $P2X^{closed}$ were subtracted from those corresponding to $P2X^{open}$. This was performed for both chains A and B. These transition vectors, $P2X^{closed}_{A} \rightarrow P2X^{open}_{A}$ and $P2X^{closed}_{B} \rightarrow P2X^{open}_{B}$, have their origin in the closed conformation and contain the information for the changes of the EV from $P2X^{closed}$ and $P2X^{open}$ for the selected chains.

4.8.3 Molecular Dynamics simulations

Standard MD simulations were performed on $P2X^{closed}_{1-ATP}$ considering the following protocol. Initially, the system was fed into the Leap module of AMBER22 where it was first neutralized. Then Na^{+} and Cl^{-} ions were incorporated to produce a 0.15M concentration. Water molecules were added to form an octahedral cell, and special care was devoted to waters molecules located inside the protein in order to conserve the shape of the receptor during the simulations. Otherwise the protein scaffold may shrink. The Amber19SB force field [**?**] was employed for computing the potential energy of the protein and water molecules while the Lipid14 force field was chosen for POPC [**?**]. The PMEMD.cuda modules of AMBER18 package was employed to run the simulations [**?**].

This initial structure was first minimized at constant volume and then heated at NVT conditions from 0 K to 100 K during 500 ps. Next, we switched to NTP conditions, and the heating was continued until 303 K was reached. Harmonic restraints of 1.5 kcal/molÅ^2 were applied on the C_α atoms of the protein in the two heating periods. This was followed by four consecutive 10.0 ns simulations at 303 K in which the restraints were gradually diminished (0.5, 0.1, 0.05 and 0.01 kcal/mol Å^2). A final equilibration stage of 200 ns was performed. The production phase consists of ten independent MD simulations started from the final structure of the equilibration stage. Each of these trajectories were run during 200 ns, conformations were sampled every 0.05 ns, and its initial velocities were randomly chosen from a Maxwellian distribution at 303K.

The SHAKE algorithm was implemented to constrain the bonds involving hydrogen atoms. Accordingly, the time step could be set to 2.0 fs. A cutoff radius of 10.0 Å was imposed to the electrostatic interactions and the Particle Mesh Ewald method was employed to calculate the long-range Coulombic force [**? ?**]. To assess the consistency of the results, we checked that each of the computed distributions presented in Fig. **??** remain unchanged when computed with each of the two halves of the whole data set.

4.9 Data Availability

The output from the Monte Carlo Markov Chain sampling of the kinetic models used to generate all figures is available via Zenodo (DOI: `https://doi.org/10.5281/zenodo.15085038`). Processed experimental data can be accessed at `https://github.com/lmoffatt/macro_dr_submission/experiments`.

4.10 Code Availability

Custom software implemented in C++20, including 11 kinetic schemes, MacroIR, MacroINR, and evidence estimation algorithms, as well as R Markdown scripts for processing MCMC outputs and reproducing figures, is available at `https://github.com/lmoffatt/macro_dr_submission`.

Data and Code Availability

All code to reproduce figures and tests is available at: `https://github.com/tu-org/macroir`. Simulation seeds and configs are archived with the manuscript.

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Competing interests

The authors declare no competing interests.

Supplementary Information (selected excerpts)